

YIMO Solutions

August 2026

Answer Keys

Div 2 Test A

1. -5 Shining
2. -1 Shining
3. 35 Karam
4. 8 Shining
5. 135 Shining
6. 219 Shining
7. 270 Karam
8. 36 Val
9. 133 Abhi
10. 8 Karam
11. 11 Rayoon
12. 90 Shining+Karam
13. 93 Neil
14. 8 Val
15. 6 Karam
16. 164 Shining
17. 559 Abhi
18. 138 Abhi
19. 42 Abhi
20. 446 Karam

Div 2 Test B

1. 999 Shining
2. 8 Shining
3. 24 Karam
4. 45 Shining
5. 7 Shining
6. 224 Shining
7. 16 Shining
8. 108 Karam
9. 40 Karam
10. 142 Abhi
11. 400 Karam
12. 311 Karam
13. 21 Karam
14. 73 Val
15. 24 Karam
16. 4 Shining
17. 112 Karam
18. 82 Karam
19. 64 Karam
20. 984 Phillip

Div 1 Test A

1. 35 Karam
2. 219 Shining
3. 270 Karam
4. 36 Val
5. 11 Rayoon
6. 90 Shining+Karam
7. 93 Neil
8. 8 Val
9. 6 Karam
10. 164 Shining
11. 174 Rayoon
12. 138 Abhi
13. 5 Karam
14. 140 Karam
15. 42 Abhi
16. 446 Karam
17. 476 Karam
18. 550 Karam
19. 60 Karam
20. 70 Karam

Div 1 Test B

1. 224 Shining
2. 108 Karam
3. 40 Karam
4. 142 Abhi
5. 400 Karam
6. 311 Karam
7. 399 Shining
8. 21 Karam
9. 73 Val
10. 151 Abhi
11. 24 Karam
12. 4 Shining
13. 112 Karam
14. 82 Karam
15. 64 Karam
16. 984 Phillip
17. 89 Karam
18. 216 Abhi
19. 754 Karam
20. 28 Neil

Solutions

Div 2 Test A

1. Since $x^2 = 25$ and $x < 0$, we have

$$\boxed{x = -5}.$$

proposed by shining

2. Subtracting $2x + 1$ from both sides of the first equation yields

$$y = -x - 1$$

and subtracting $3x + 2$ from both sides of the second equation yields

$$y = -2x - 2$$

so

$$-x - 1 = -2x - 2$$

adding 2 to both sides gives

$$x - 1 = -2$$

and adding 1 to both sides we get

$$x = -1$$

. Substituting our value for x back into the first equation to solve for y gives us

$$y = -(-1) - 1 = 1 - 1 = 0$$

so (a, b) is $(-1, 0)$ and we get

$$a + b = -1 + 0 = \boxed{-1}.$$

proposed by shining.

3. Triangle BDC is isosceles. Thus

$$\angle DBC = \angle BCD = 40^\circ.$$

Therefore,

$$\angle ABD = 75^\circ - 40^\circ = \boxed{35^\circ}.$$

proposed by karam.

4. The two-digit factors of 64 are $2^4 = 16$, $2^5 = 32$, $2^6 = 64$ each of which is a multiple of 3. Thus there are 3 two-digit multiples of 8 that are also divisors of 64. The smallest two-digit multiple of 8 is 16 and the largest is 96 so we have

$$\frac{96 - 16}{8} + 1 = 11$$

two-digit multiples of 8. Subtracting off the 3 divisors of 64 gives us

$$11 - 3 = \boxed{8}.$$

proposed by shining.

5. Assuming the days are independent, there are 7 choices for the one day without rain. Hence

$$P(6 \text{ rainy days}) = \binom{7}{6} \left(\frac{1}{2}\right)^7 = \frac{7}{128}.$$

Therefore,

$$m + n = \boxed{135}.$$

proposed by shining.

6. Let $y = 2^x$ we express every term in terms of y .

- $8^x = (2^3)^x = (2^x)^3 = y^3$
- $4^x = (2^2)^x = (2^x)^2 = y^2$
- $4^{x+1} = 4 \cdot 4^x = 4y^2$
- $2^{x+1} = 2 \cdot 2^x = 2y$

Substituting into the original equation gives

$$y^3 + y^2 = 4y^2 + 2y \implies y^3 - 3y^2 - 2y = 0 \implies y(y^2 - 3y - 2) = 0 \implies y^2 - 3y - 2 = 0$$

. We solve for y using quadratic formula and get

$$y = \frac{3 \pm \sqrt{(-3)^2 - 4(1)(-2)}}{2} = \frac{3 \pm \sqrt{17}}{2}$$

we take the positive root as $y = 2^x > 0$ so

$$y = \frac{3 + \sqrt{17}}{2}.$$

Notice that

$$4^{2x} = (2^2)^{2x} = (2^x)^4 = y^4$$

so

$$y^2 = \left(\frac{3 + \sqrt{17}}{2} \right)^2 = \frac{9 + 6\sqrt{17} + 17}{4} = \frac{26 + 6\sqrt{17}}{4} = \frac{13 + 3\sqrt{17}}{2}.$$

We square again to get

$$y^4 = \left(\frac{13 + 3\sqrt{17}}{2} \right)^2 = \frac{169 + 78\sqrt{17} + 9(17)}{4} = \frac{169 + 78\sqrt{17} + 153}{4} = \frac{322 + 78\sqrt{17}}{4}.$$

Next we simplify the fraction to arrive at

$$4^{2x} = \frac{161 + 39\sqrt{17}}{2} \implies a + b + c + d = 161 + 39 + 17 + 2 = \boxed{219}.$$

proposed by shining

7. There are

$$\binom{27}{2} = 351$$

pairs of blocks. A laser can hit both blocks only if they lie in the same row.

There are $3 \cdot 3^2 = 27$ rows, and each row contains $\binom{3}{2} = 3$ pairs. Thus the answer is

$$351 - 27 \binom{3}{2} = \boxed{270}.$$

proposed by karam.

8. Let $d(n)$ be the number of positive divisors of n . The condition states

$$d(n) \geq \frac{n}{2}$$

Because divisors pair up, the number of divisors cannot exceed $2\sqrt{n}$ (because every divisor d from $1, 2, \dots, \sqrt{n}$ is paired up with $\frac{n}{d}$ from the set $\sqrt{n} + 1, \dots, n$) so we get

$$d(n) \leq 2\sqrt{n}$$

Combining these two gives

$$\frac{n}{2} \leq 2\sqrt{n} \implies n \leq 4\sqrt{n} \implies n^2 \leq 16n \implies n \leq 16$$

We test all integers $1 \leq n \leq 16$ to see if $d(n) \geq n/2$:

- $n = 1$: Divisors are $\{1\}$. So $d(1) = 1$, which is ≥ 0.5 (Valid)
- $n = 2$: Divisors are $\{1, 2\}$. So $d(2) = 2$, which is ≥ 1 (Valid)
- $n = 3$: Divisors are $\{1, 3\}$. So $d(3) = 2$, which is ≥ 1.5 (Valid)
- $n = 4$: Divisors are $\{1, 2, 4\}$. So $d(4) = 3$, which is ≥ 2 (Valid)
- $n = 5$: Divisors are $\{1, 5\}$. So $d(5) = 2$, which is < 2.5 (Invalid)
- $n = 6$: Divisors are $\{1, 2, 3, 6\}$. So $d(6) = 4$, which is ≥ 3 (Valid)
- $n = 7$: Divisors are $\{1, 7\}$. So $d(7) = 2$, which is < 3.5 (Invalid)
- $n = 8$: Divisors are $\{1, 2, 4, 8\}$. So $d(8) = 4$, which is ≥ 4 (Valid)
- $n = 9$: Divisors are $\{1, 3, 9\}$. So $d(9) = 3$, which is < 4.5 (Invalid)
- $n = 10$: Divisors are $\{1, 2, 5, 10\}$. So $d(10) = 4$, which is < 5 (Invalid)
- $n = 11$: Divisors are $\{1, 11\}$. So $d(11) = 2$, which is < 5.5 (Invalid)
- $n = 12$: Divisors are $\{1, 2, 3, 4, 6, 12\}$. So $d(12) = 6$, which is ≥ 6 (Valid)
- $n = 13$: Divisors are $\{1, 13\}$. So $d(13) = 2$, which is < 6.5 (Invalid)
- $n = 14$: Divisors are $\{1, 2, 7, 14\}$. So $d(14) = 4$, which is < 7 (Invalid)
- $n = 15$: Divisors are $\{1, 3, 5, 15\}$. So $d(15) = 4$, which is < 7.5 (Invalid)
- $n = 16$: Divisors are $\{1, 2, 4, 8, 16\}$. So $d(16) = 5$, which is < 8 (Invalid)

The valid n values are 1, 2, 3, 4, 6, 8, and 12. Summing these values together gives

$$1 + 2 + 3 + 4 + 6 + 8 + 12 = \boxed{36}$$

proposed by val.

9. Since

$$5^2 + 12^2 = 13^2,$$

triangle O_1XO_2 is right at X . Its altitude to O_1O_2 has length

$$\frac{5 \cdot 12}{13} = \frac{60}{13}.$$

This altitude is half of the common chord XY , so

$$XY = \frac{120}{13}.$$

Therefore,

$$m + n = \boxed{133}.$$

proposed by abhi.

10. By Vieta's formulas, the sum of the roots of the polynomial is

$$a + b + c = 6$$

. We rewrite each factor in the expression using this sum

- $a + b - c = (a + b + c) - 2c = 6 - 2c$
- $b + c - a = (a + b + c) - 2a = 6 - 2a$
- $c + a - b = (a + b + c) - 2b = 6 - 2b$

Substituting these back gives us

$$|(6 - 2a)(6 - 2b)(6 - 2c)|$$

Factoring out 2 from each of the three terms gives

$$|2(3 - a) \cdot 2(3 - b) \cdot 2(3 - c)| = 8|(3 - a)(3 - b)(3 - c)|$$

Since a, b , and c are the roots of the polynomial $P(x) = x^3 - 6x^2 + 11x - 7$, it can be written in factored form as the following

$$P(x) = (x - a)(x - b)(x - c)$$

If we evaluate $P(x)$ at $x = 3$ we get

$$P(3) = (3 - a)(3 - b)(3 - c)$$

We calculate $P(3)$

$$P(3) = 3^3 - 6(3)^2 + 11(3) - 7$$

$$P(3) = 27 - 54 + 33 - 7 = -1$$

Substituting $P(3) = -1$ back into our expression gives

$$8|-1| = 8(1) = \boxed{8}$$

proposed by karam.

11. One of the ten dice is a fair 10sided die. Fix the results of the other nine dice, and let their sum be S . We need the 10sided die to show a number r such that

$$S + r \equiv 1 \pmod{10}.$$

The possible rolls $1, 2, \dots, 10$ represent every residue modulo 10 exactly once. Regardless of the value of S , exactly one of the ten possible rolls makes the total have units digit 1.

Hence

$$P(\text{units digit } 1) = \frac{1}{10}.$$

So the answer is

$$m + n = 1 + 10 = \boxed{11}.$$

proposed by Rayoon

12. Let x be a positive integer, we want to find the number of perfect cubes $x^3 < 10^6$ that cannot be written in the form y^2 for any integer y . First, we find the total number of positive perfect cubes less than 10^6

$$x^3 < 10^6 \implies x < (10^6)^{1/3} \implies x < 100$$

Since x is a positive integer, x can range from 1 to 99 giving us a total of 99 perfect cubes. Next, a number is both a perfect cube and a perfect square if and only if it is a perfect sixth power m^6 . We count how many of our 99 perfect cubes are also perfect squares. This happens when the base x itself is a perfect square $x = m^2$ which means

$$m^2 < 100 \implies m < 10$$

The possible positive integer values for m are 1, 2, 3, 4, 5, 6, 7, 8, and 9 thus there are exactly 9 perfect cubes that are also perfect squares. To find the perfect cubes that are not perfect squares, we subtract the overlapping sixth powers from the total number of cubes to arrive at our final answer

$$\text{Total Cubes} - \text{Cubes that are Squares} = 99 - 9 = \boxed{90}$$

proposed by Shining+Karam

13. Let

$$N = 2026^{2026}.$$

Since

$$\begin{aligned} 2026 &= 2 \cdot 1013 \\ N &= 2^{2026} \cdot 1013^{2026}. \end{aligned}$$

Every positive divisor of N has the form

$$2^a 1013^b, \quad 0 \leq a, b \leq 2026.$$

The total number of positive divisors is

$$(2026 + 1)^2 = 2027^2 = 4,108,729.$$

$$\sqrt{N} = 2026^{1013}.$$

Every divisor $d \neq \sqrt{N}$ can be paired with N/d . Exactly one divisor in each pair is less than $\sqrt{N}$. Since $\sqrt{N}$ is the only unpaired divisor, the number of divisors less than $\sqrt{N}$ is

$$\frac{4,108,729 - 1}{2} = 2,054,364.$$

Desired probability is

$$\frac{2,054,364}{4,108,729}.$$

So only adding the last 3 digits,

$$m + n = 364 + 729 = 1093$$

The last three digits are

$$\boxed{093}.$$

proposed by Neil

14. Let N be the maximum number of players, and let the challenges be numbered $1, 2, \dots, 8$, each player chooses a subset of 3 distinct challenges. First, we determine the maximum number of times any single

challenge can be chosen across all players. Suppose a specific challenge is chosen by k players. Because no two players can share a second challenge in common, the other 2 challenges chosen by each of these k players must all be distinct. The total number of remaining challenges required by these k players is $2k$. Since there are only 7 other challenges available in the entire pool, we must have

$$2k \leq 7 \implies k \leq 3.5$$

Since k must be an integer, any single challenge can appear in at most 3 players' selections. Next, we bound the total number of players. Since there are 8 challenges and each can be used a maximum of 3 times, there are at most $8 \times 3 = 24$ available slots in total. Since each player must take exactly 3 challenges, the maximum number of players is

$$N \leq \frac{24}{3} = 8$$

To prove that 8 players is achievable, we construct a valid arrangement.

- Player 1: $\{8, 1, 2\}$
- Player 2: $\{8, 3, 4\}$
- Player 3: $\{8, 5, 6\}$
- Player 4: $\{7, 1, 3\}$
- Player 5: $\{7, 2, 5\}$
- Player 6: $\{7, 4, 6\}$
- Player 7: $\{1, 4, 5\}$
- Player 8: $\{2, 3, 6\}$

Checking the intersections confirms that no two players share more than 1 challenge, thus, the maximum number of players is $\boxed{8}$.

proposed by Val

15. Let

$$d = \gcd(m, n), \quad m = da, \quad n = db,$$

where $\gcd(a, b) = 1$. Then

$$\text{lcm}(m, n) = dab.$$

Substituting into the given equation gives

$$dab = da + db + d + 6,$$

so

$$d(ab - a - b - 1) = 6.$$

Set

$$e = ab - a - b - 1.$$

Since $de = 6$ and we are dealing with integers,

$$e \in \{1, 2, 3, 6\} \quad \text{and} \quad d = \frac{6}{e}.$$

The equation for a and b can be rewritten as

$$(a - 1)(b - 1) = e + 2.$$

We now consider the four possibilities. Casework: If $e = 1$, then

$$(a - 1)(b - 1) = 3.$$

This gives $(a, b) = (2, 4)$ or $(4, 2)$, neither of which is coprime (we are looking for coprime numbers a and b because we assume d is the only common divisor and a or b does not divide the other).

If $e = 2$, then

$$(a - 1)(b - 1) = 4.$$

The coprime possibilities are

$$(a, b) = (2, 5), (5, 2).$$

Since $d = 3$, these give

$$(m, n) = (6, 15), (15, 6).$$

If $e = 3$, then

$$(a - 1)(b - 1) = 5.$$

This gives $(a, b) = (2, 6)$ or $(6, 2)$, neither of which is coprime.

If $e = 6$, then

$$(a - 1)(b - 1) = 8.$$

The coprime possibilities are

$$(a, b) = (2, 9), (3, 5), (5, 3), (9, 2).$$

Since $d = 1$, these give the same four ordered pairs for (m, n) .

Therefore, the six ordered pairs are

$$(6, 15), (15, 6), (2, 9), (3, 5), (5, 3), (9, 2).$$

All satisfy the given bounds, so the answer is

$$\boxed{6}.$$

proposed by Karam

16. We can rewrite the floor function term as

$$\left\lfloor \frac{i^2}{7} \right\rfloor = \frac{i^2 - (i^2 \bmod 7)}{7}$$

Splitting the original summation into two components gives

$$S = \frac{1}{7} \sum_{i=1}^{2026} i^2 - \frac{1}{7} \sum_{i=1}^{2026} (i^2 \bmod 7)$$

Let the first part be $T_1 = \frac{1}{7} \sum_{i=1}^{2026} i^2$ using the formula for the sum of squares yields

$$T_1 = \frac{1}{7} \cdot \frac{2026(2027)(4053)}{6} = \frac{2026 \times 2027 \times 579}{6}$$

$$T_1 = 1013 \times 2027 \times 193$$

Since we only need the final 3 digits, we evaluate $T_1 \pmod{1000}$

$$T_1 \equiv 13 \times 27 \times 193 \pmod{1000}$$

$$T_1 \equiv 351 \times 193 = 67743 \equiv 743 \pmod{1000}$$

Let the second part be $T_2 = \frac{1}{7} \sum_{i=1}^{2026} (i^2 \bmod 7)$. The remainders of $i^2 \pmod{7}$ repeat in a period of 7

$$\{1, 4, 2, 2, 4, 1, 0\}$$

The sum of a single full period is $1 + 4 + 2 + 2 + 4 + 1 + 0 = 14$. Dividing the total by the period length 7 gives

$$2026 = 289 \times 7 + 3$$

The sum consists of 289 complete periods and the first 3 terms of the next period $(1, 4, 2)$ which sums to

$$\sum_{i=1}^{2026} (i^2 \bmod 7) = (289 \times 14) + (1 + 4 + 2) = 4046 + 7 = 4053$$

$$T_2 = \frac{4053}{7} = 579$$

Subtracting T_2 from T_1 to determine the final value modulo 1000 gives

$$S \equiv 743 - 579 = \boxed{164} \pmod{1000}$$

proposed by Shining

17. Let

$$x = AC, \quad y = BD.$$

Because $ABCD$ is cyclic, opposite angles are supplementary. Thus

$$\cos \angle ABC = -\cos \angle ADC.$$

Applying the Law of Cosines in triangles ABC and ADC , we get

$$\cos \angle ABC = \frac{3^2 + 4^2 - x^2}{2 \cdot 3 \cdot 4} = \frac{25 - x^2}{24}$$

and

$$\cos \angle ADC = \frac{6^2 + 5^2 - x^2}{2 \cdot 6 \cdot 5} = \frac{61 - x^2}{60}.$$

Therefore,

$$\frac{25 - x^2}{24} = -\frac{61 - x^2}{60}.$$

Multiplying by 120 gives

$$5(25 - x^2) = -2(61 - x^2),$$

so

$$x^2 = \frac{247}{7}.$$

Similarly,

$$\cos \angle BAD = -\cos \angle BCD.$$

Applying the Law of Cosines again gives

$$\frac{45 - y^2}{36} = -\frac{41 - y^2}{40}.$$

Solving,

$$y^2 = \frac{819}{19}.$$

Hence

$$AC^2 + BD^2 = \frac{247}{7} + \frac{819}{19} = \frac{4693 + 5733}{133} = \frac{10426}{133}.$$

Adding $m+n$ and taking the last 3 digits,

$$m + n = 10426 + 133 = \boxed{559}.$$

Proposed by Abhi

18. Let $N = \frac{2026!}{11^a \cdot 13^b}$. Since $143 = 11 \times 13$, we can solve for $N \pmod{11}$ and $N \pmod{13}$ separately, then combine them using the Chinese Remainder Theorem. Using Legendre's Formula we can compute the exponents,

$$a = \left\lfloor \frac{2026}{11} \right\rfloor + \left\lfloor \frac{2026}{121} \right\rfloor + \left\lfloor \frac{2026}{1331} \right\rfloor = 184 + 16 + 1 = 201$$

$$b = \left\lfloor \frac{2026}{13} \right\rfloor + \left\lfloor \frac{2026}{169} \right\rfloor = 155 + 11 = 166$$

We isolate the prime 11 component from the rest of the expression:

$$N = \frac{2026!}{11^{201}} \times \frac{1}{13^{166}} \pmod{11}$$

First, we evaluate the factorial part using the generalization of Wilson's Theorem, writing 2026 in base 11 gives digits 1, 5, 8, 2 so

$$\frac{2026!}{11^{201}} \equiv (-1)^{201} \times 1! \times 5! \times 8! \times 2! \pmod{11}$$

- $(-1)^{201} = -1$
- $5! = 120 \equiv -1 \pmod{11}$
- $8! \equiv 5 \pmod{11}$
- $2! = 2$

$$\frac{2026!}{11^{201}} \equiv (-1) \times 1 \times (-1) \times 5 \times 2 = 10 \pmod{11}$$

Second, we evaluate the division by 13^{166} . Since $13 \equiv 2 \pmod{11}$, by Fermat's Little Theorem we have

$$2^{166} \equiv 2^6 = 64 \equiv 9 \pmod{11}$$

The modular inverse of $9 \pmod{11}$ is 5 so

$$N \equiv 10 \times 5 = 50 \equiv 6 \pmod{11}$$

We rewrite the expression isolating the prime 13 component to get

$$N = \frac{2026!}{13^{166}} \times \frac{1}{11^{201}} \pmod{13}$$

Converting 2026 to base 13 gives digits 11, 12, 11 so

$$\frac{2026!}{13^{166}} \equiv (-1)^{166} \times 11! \times 12! \times 11! \pmod{13}$$

- $(-1)^{166} = 1$
- $12! \equiv -1 \pmod{13}$

- $11! \equiv 1 \pmod{13}$

$$\frac{2026!}{13^{166}} \equiv 1 \times 1 \times (-1) \times 1 = -1 \equiv 12 \pmod{13}$$

Next, we evaluate the division by 11^{201} , since $11 \equiv -2 \pmod{13}$ we have

$$(-2)^{201} = -2^{201} \equiv -2^9 = -512 \equiv -5 \equiv 8 \pmod{13}$$

The modular inverse of 8 $\pmod{13}$ is 5 so

$$N \equiv 12 \times 5 = 60 \equiv 8 \pmod{13}$$

We now have a system of congruences

- $N \equiv 6 \pmod{11}$
- $N \equiv 8 \pmod{13}$

From the second congruence, $N = 13k + 8$ so

$$13k + 8 \equiv 6 \pmod{11}$$

$$2k \equiv -2 \equiv 9 \equiv 20 \pmod{11} \implies k \equiv 10 \pmod{11}$$

Substituting $k = 10$ back into the formula for N gives us our final answer of

$$N = 13(10) + 8 = \boxed{138}.$$

Proposed by Abhi

19. Suppose the roster contains k knights and k rabbits. The players can be selected in

$$\binom{9}{k} \binom{11}{k}$$

ways.

After the players are chosen, there are k choices for the knight captain and k choices for the rabbit captain. Thus, for a fixed value of k , the number of rosters is

$$k^2 \binom{9}{k} \binom{11}{k}.$$

Since there are only 9 knights,

$$N = \sum_{k=1}^9 k^2 \binom{9}{k} \binom{11}{k}.$$

Using the absorption identity

$$k \binom{9}{k} = 9 \binom{8}{k-1} \quad \text{and} \quad k \binom{11}{k} = 11 \binom{10}{k-1},$$

we obtain

$$N = 99 \sum_{k=1}^9 \binom{8}{k-1} \binom{10}{k-1}.$$

Let $j = k - 1$. Then

$$N = 99 \sum_{j=0}^8 \binom{8}{j} \binom{10}{j}.$$

Since

$$\binom{10}{j} = \binom{10}{10-j},$$

Vandermonde's identity gives

$$\sum_{j=0}^8 \binom{8}{j} \binom{10}{10-j} = \binom{18}{10} = \binom{18}{8}.$$

$$N = 99 \binom{18}{8} = 99(43758) = 4,332,042.$$

The last three digits are

$$\boxed{042}.$$

Proposed by Abhi

20. We expand both base representations to convert them into base 10

$$N = 1 \cdot b^2 + 2 \cdot b + 3 = b^2 + 2b + 3$$

$$N = 3 \cdot c^2 + 2 \cdot c + 1 = 3c^2 + 2c + 1$$

Equating the two gives

$$b^2 + 2b + 3 = 3c^2 + 2c + 1 \implies b^2 + 2b + 2 = 3c^2 + 2c$$

Completing the square on the LHS yields

$$(b+1)^2 + 1 = 3c^2 + 2c$$

Multiply the equation by 3 and complete the square to get

$$3(b+1)^2 + 3 = 9c^2 + 6c$$

$$3(b+1)^2 + 4 = (3c+1)^2$$

Rearranging this into a standard Pell-like form gives

$$(3c+1)^2 - 3(b+1)^2 = 4$$

Let $X = 3c + 1$ and $Y = b + 1$. The equation simplifies to

$$X^2 - 3Y^2 = 4$$

Since the digits 3 and 2 are used in the representations, the bases must satisfy $b > 3$ and $c > 3$. The positive integer solutions (X_n, Y_n) can be generated from the solution $(2, 0)$ via the linear recurrences

$$X_{n+1} = 2X_n + 3Y_n$$

$$Y_{n+1} = X_n + 2Y_n$$

We now simply check the solutions

- $(X_0, Y_0) = (2, 0) \implies b = -1, c = 1/3$ (Invalid)
- $(X_1, Y_1) = (4, 2) \implies b = 1, c = 1$ (Invalid)

- $(X_2, Y_2) = (14, 8) \implies 3c + 1 = 14 \implies c = 13/3$ (Non-integer)
- $(X_3, Y_3) = (52, 30) \implies 3c + 1 = 52 \implies c = 17$ and $b + 1 = 30 \implies b = 29$ (Valid)
- $(X_4, Y_4) = (194, 112) \implies 3c + 1 = 194 \implies c = 193/3$ (Non-integer)
- $(X_5, Y_5) = (724, 418) \implies 3c + 1 = 724 \implies c = 241$ and $b + 1 = 418 \implies b = 417$ (Valid)
- $(X_6, Y_6) = (2702, 1560) \implies 3c + 1 = 2702 \implies c = 2701/3$ (Non-integer)
- $(X_7, Y_7) = (10084, 5822) \implies b + 1 = 5822 \implies b = 5821$ (Invalid, exceeds 1000)

The valid values for b less than 1000 are 29 and 417 which gives

$$\text{Sum} = 29 + 417 = \boxed{446}$$

Proposed by Karam

Solutions to odd problem numbers (1,3,5,7,...19) written by Abhi, even (2,4,6,8...,20) by Karam.

Div 2 Test B

1. The even positive integers less than 2000 are

$$2, 4, 6, \dots, 1998.$$

This is an arithmetic sequence with first term 2, last term 1998, and common difference 2.

The number of terms is:

$$\frac{1998 - 2}{2} + 1 = \frac{1996}{2} + 1 = 998 + 1 = 999.$$

Therefore, the number of even positive integers less than 2000 is

$$\boxed{999}$$

proposed by Shining

2. The cleanest way to handle this is to add the circles one at a time and ask how much each new circle buys us.

One circle splits the plane into an inside and an outside: 2 regions.

Now drop in a second circle. Two distinct circles meet in at most 2 points, and those 2 points cut the new circle into 2 arcs. Here is the key observation, and it is worth saying slowly: *every arc of the new circle slices some old region in two*, so each arc contributes exactly one new region. Two arcs, two new regions, and we are at 4.

The third circle can meet each of the previous two in 2 points, for 4 intersection points in total. Those 4 points chop the third circle into 4 arcs, so we gain 4 more regions:

$$2 + 2 + 4 = 8.$$

We should make sure this is actually achievable and not just an upper bound — the count above assumed all intersections happen and are distinct. Take three circles of equal radius whose centers form a small triangle (the classic Venn diagram picture). Every pair crosses twice, no three circles share a point, and you can count the eight regions directly: the outside, three “only” regions, three “exactly two” lens regions, and the middle where all three overlap. That is $1 + 3 + 3 + 1 = 8$. $\boxed{8}$

Proposed by Shining

3. After 40 minutes, LeBron has traveled $8 \cdot \frac{40}{60} = \frac{16}{3}$ miles. Jordan has traveled $5 \cdot \frac{40}{60} = \frac{10}{3}$ miles. Therefore, LeBron is ahead by $\frac{16}{3} - \frac{10}{3} = 2$ miles. While LeBron waits, Jordan continues traveling at 5 miles per hour. Thus, Jordan closes the 2-mile gap in $\frac{2}{5}$ hours = $\frac{2}{5} \times 60 = 24$ minutes. Hence, LeBron must wait

$$\boxed{24}$$

minutes.

proposed by Karam

4. The instinct here should be that $f(x) \geq x$ is an extremely demanding condition, because digit sums are tiny compared to the numbers they come from. Let us make that precise.

Write x with d digits, $x = \overline{a_{d-1}a_{d-2}\cdots a_0}$, so

$$x = \sum_{i=0}^{d-1} a_i 10^i, \quad f(x) = \sum_{i=0}^{d-1} a_i.$$

Subtracting,

$$x - f(x) = \sum_{i=0}^{d-1} a_i (10^i - 1).$$

Every term on the right is nonnegative, since $10^i \geq 1$. So $x \geq f(x)$ *always*, and the inequality $f(x) \geq x$ can only hold when it is an equality.

Equality forces every term $a_i(10^i - 1)$ to vanish. The $i = 0$ term vanishes automatically, but for $i \geq 1$ we have $10^i - 1 > 0$, so we need $a_i = 0$ for all $i \geq 1$. In other words x has a single (nonzero) digit:

$$x \in \{1, 2, \dots, 9\}.$$

These all genuinely work, since a one-digit number is its own digit sum. Summing,

$$1 + 2 + \dots + 9 = \frac{9 \cdot 10}{2} = \boxed{45}.$$

Proposed by Shining

5. Factor the expression into $x^2 - 8x + 12 = (x - 6)(x - 2)$. For the product $(x - 6)(x - 2)$ to be a positive prime, one of the factors must be 1 or -1 . There are two possibilities if one of the factors is 1: $x - 6 = 1 \implies x = 7$ which gives 5, a prime. Or, $x - 2 = 1 \implies x = 3$ which gives -3 , which is not positive. If one of the factors is -1 : $x - 6 = -1 \implies x = 5$ which gives -3 , which is not positive. Or, $x - 2 = -1 \implies x = 1$, which gives 5 but this is not the maximum. Hence, the maximum positive integer satisfying the condition is

$$\boxed{7}$$

proposed by Shining.

6. The expression is built up by nesting, so we peel it off one layer at a time rather than expanding anything. Start from the outside:

$$((x + 1)^2 + 1)^2 + 1 = 290 \implies ((x + 1)^2 + 1)^2 = 289.$$

Taking square roots gives $(x + 1)^2 + 1 = \pm 17$, but $(x + 1)^2 + 1 \geq 1 > 0$, so we keep only the positive root:

$$(x + 1)^2 + 1 = 17 \implies (x + 1)^2 = 16 \implies x + 1 = \pm 4.$$

That leaves $x = 3$ or $x = -5$. The condition $x < 0$ is exactly what picks between them (this is why the problem included it), so $x = -5$.

Now evaluate the second expression at $x = -5$, again from the inside out:

$$(x - 1)^2 = (-6)^2 = 36, \quad 36 - 1 = 35, \quad 35^2 = 1225, \quad 1225 - 1 = 1224.$$

The last three digits are $\boxed{224}$.

Proposed by Shining

7. A regular octagon has $\frac{8(8-3)}{2} = 20$ diagonals. A diagonal passes through the center if and only if it joins a pair of opposite vertices. Since an octagon has 4 pairs of opposite vertices, there are 4 such diagonals. Hence, the number of diagonals that do not pass through the center is $20 - 4 = 16$. Therefore, the answer is

$$\boxed{16}$$

proposed by Shining.

8. Setting up coordinates. The sheet has area 144, so it is 12×12 ; put it at $[0, 12]^2$. Fold along $y = 6$ to get the rectangle $[0, 12] \times [0, 6]$, then fold along $x = 6$ to get the square $[0, 6]^2$. Track the center of the original sheet, $(6, 6)$: after both folds it sits at the point $(6, 6)$, which is a *corner* of the folded square. Better still, the two edges meeting at that corner are precisely the two creases. This is the whole reason the problem works, and it also tells us which corner gets cut — if we snipped any other corner the holes would not land in the center.

What the cut unfolds into. A straight cut removing the corner $(6, 6)$ must meet the two edges through that corner, say at $(6 - a, 6)$ and $(6, 6 - b)$ with $a, b > 0$. It slices through all four layers. Unfolding reflects that little right triangle across both creases, so the hole is the quadrilateral with vertices

$$(6 \pm a, 6) \quad \text{and} \quad (6, 6 \pm b),$$

a rhombus centered at $(6, 6)$ with diagonals $2a$ and $2b$. Each of its four sides is a copy of the cut, of length $\sqrt{a^2 + b^2}$.

Using the perimeter. The perimeter condition says

$$4\sqrt{a^2 + b^2} = 24 \implies a^2 + b^2 = 36.$$

The hole's area, from the diagonal formula for a rhombus, is

$$[\text{hole}] = \frac{1}{2}(2a)(2b) = 2ab.$$

By AM-GM, $2ab \leq a^2 + b^2 = 36$, with equality when $a = b = 3\sqrt{2}$ (which comfortably fits inside the 6×6 square). So the hole can be as large as 36, and the paper left behind is

$$144 - 36 = \boxed{108}.$$

Proposed by Karam

9. Since $x - 2 \geq 0$, we must have $x \geq 2$. Move one radical to the other side to get $\sqrt{2x + 5} = 3 + \sqrt{x - 2}$. Then, square both sides to get $x - 2 = 6\sqrt{x - 2}$. Let $y = \sqrt{x - 2} \geq 0$. Then $y^2 = 6y$ so $y(y - 6) = 0$. Hence, $y = 0 \implies x = 2$, or $y = 6 \implies x = 38$. Therefore, the sum of all solutions is $2 + 38 = 40$.

$$\boxed{40}$$

proposed by Karam

10. Write the number as $d_1 d_2 d_3 d_4 d_5$. The conditions are

$$d_1 < d_2 < d_3 \quad \text{and} \quad d_3 > d_4 > d_5.$$

Notice that d_3 appears in both chains — it is the peak of the number — so the natural move is to condition on it. Once $d_3 = k$ is fixed, the left half and the right half are chosen completely independently.

The left half. We need $d_1 < d_2 < k$, and since d_1 is the leading digit it cannot be 0. So d_1, d_2 is any 2-element subset of $\{1, 2, \dots, k - 1\}$ (a set determines the order, since the digits must increase). That is $\binom{k-1}{2}$ ways.

The right half. We need $k > d_4 > d_5$, and here 0 is allowed, so d_4, d_5 is any 2-element subset of $\{0, 1, \dots, k - 1\}$: that is $\binom{k}{2}$ ways.

The asymmetry between the two halves is exactly the leading-digit restriction, and it is the easiest thing to lose. Multiplying and summing over the peak k (which must be at least 3 for the left half to be possible):

$$N = \sum_{k=3}^9 \binom{k-1}{2} \binom{k}{2}.$$

Tabulating:

k	$\binom{k-1}{2}$	$\binom{k}{2}$	product
3	1	3	3
4	3	6	18
5	6	10	60
6	10	15	150
7	15	21	315
8	21	28	588
9	28	36	1008

Adding these up:

$$3 + 18 + 60 + 150 + 315 + 588 + 1008 = 2142.$$

So $N = 2142$, and its last three digits are $\boxed{142}$.

Proposed by Abhi

11. Let $a = xy$, $b = yz$, $c = zx$. Then the system becomes

$$\begin{aligned} a + c &= 125, \\ a + b &= 89, \\ b + c &= 164. \end{aligned}$$

Adding the three equations gives $2(a + b + c) = 125 + 89 + 164 = 378$, so $a + b + c = 189$. Hence, $a = 189 - 164 = 25$, $b = 189 - 125 = 64$, $c = 189 - 89 = 100$. So, $xy = 25$, $yz = 64$, $zx = 100$. Multiplying these equations gives $(xy)(yz)(zx) = x^2y^2z^2 = (xyz)^2 = 25 \cdot 64 \cdot 100 = 160000$. Since $x, y, z > 0$,

$$xyz = \sqrt{160000} = \boxed{400}$$

Proposed by Karam

12. Let the 10 identical packages be distributed into 4 distinct mailboxes. By the Stars and Bars, the total number of equally likely ways to distribute the packages is

$$\text{Total Ways} = \binom{10 + 4 - 1}{4 - 1} = \binom{13}{3} = \frac{13 \times 12 \times 11}{3 \times 2 \times 1} = 286$$

By Linearity of Expectation, the expected number of overloaded mailboxes E is equal to 4 times the probability that a specific mailbox is overloaded namely

$$E = 4 \times P(\text{Mailbox 1 contains } \geq 4 \text{ packages})$$

To find the number of ways a specific mailbox receives 4 or more packages, we pre-allocate 4 packages to that mailbox. We then distribute the remaining $10 - 4 = 6$ packages across all 4 mailboxes without restriction

$$\text{Ways to Overload} = \binom{6 + 4 - 1}{4 - 1} = \binom{9}{3} = \frac{9 \times 8 \times 7}{3 \times 2 \times 1} = 84$$

So probability that a specific mailbox is overloaded is:

$$P = \frac{84}{286} = \frac{42}{143}$$

Now, we calculate the expected value E to be

$$E = 4 \times \frac{42}{143} = \frac{168}{143} \implies m + n = 168 + 143 = \boxed{311}$$

Proposed by Karam.

13. Since $g(x) \geq 0$, solutions can only occur where $\sin(10x) \geq 0$. On $[0, \pi]$, $\sin(10x)$ is positive on the five intervals

$$\left(0, \frac{\pi}{10}\right), \left(\frac{\pi}{5}, \frac{3\pi}{10}\right), \left(\frac{2\pi}{5}, \frac{\pi}{2}\right), \left(\frac{3\pi}{5}, \frac{7\pi}{10}\right), \left(\frac{4\pi}{5}, \frac{9\pi}{10}\right).$$

At each endpoint, $f(x) = 0 < g(x)$, while at the midpoint of each interval, $f(x) = 1 > g(x)$. Since $f(x) - g(x)$ is continuous and changes sign exactly once on each half of every positive interval, each interval contributes exactly two solutions. Thus, there are 10 solutions in $(0, \pi)$, together with the endpoint $x = \pi$, giving 11 solutions on $[0, \pi]$. Similarly, on $[-\pi, 0]$, there are four full positive intervals contributing 8 solutions and the partial interval $[-\pi, -9\pi/10]$, which contributes one interior solution together with the endpoint $x = -\pi$, for a total of 10 solutions.

$$11 + 10 = \boxed{21}$$

proposed by Karam

14. As always with these games, we hunt for the *losing positions*: the values of n such that the player *about to move* with n coins loses under optimal play. Call these L -positions.

By definition $n = 1$ is an L -position. Let us compute a few more by hand, remembering that from n coins you may leave anywhere from $\lceil n/2 \rceil$ to $n - 1$ coins.

- $n = 2$: take 1, leaving your opponent at 1. Win.
- $n = 3$: you may only take 1 (half of 3 rounds down), leaving 2 — a winning position for your opponent. Lose.
- $n = 4, 5, 6$: you can leave exactly 3 (take 1, 2, or 3 respectively, all within the half-limit). Win.
- $n = 7$: you may take at most 3, so you leave 4, 5, or 6 — all winners for your opponent. Lose.

The pattern 1, 3, 7, ... is unmistakable: the L -positions are $n = 2^k - 1$. Here is the proof, and both halves matter.

From $2^k - 1$ you are stuck. You may leave anything in $[2^{k-1}, 2^k - 2]$. The largest number of the form $2^j - 1$ in that window is $2^{k-1} - 1$, which is strictly below it, so every legal move lands you strictly between two consecutive L -positions — never on one.

From any other n you can escape. Let $2^k - 1$ be the largest such number less than n , so $2^k - 1 < n < 2^{k+1} - 1$, i.e. $n \leq 2^{k+1} - 2$. You want to remove $n - (2^k - 1)$ coins, which is at least 1; it is legal provided

$$n - (2^k - 1) \leq \frac{n}{2} \iff \frac{n}{2} \leq 2^k - 1 \iff n \leq 2^{k+1} - 2,$$

which is exactly the range we are in. So the move to $2^k - 1$ is always available.

Therefore the L -positions are precisely 1, 3, 7, 15, 31, 63, 127, 255, ..., and Shakira's job is to hand Pitbull one of them. Since $127 < 200 < 255$, she should leave 127 coins, which means removing

$$200 - 127 = \boxed{73}$$

coins. This is legal because $73 \leq 100$. From there she simply keeps returning Pitbull to 63, 31, 15, 7, 3, and finally 1.

Proposed by Val.

15. Since $7 \mid d$, we must have $7 \mid n$ and $7 \mid 2^n - 1$. The order of 2 modulo 7 is 3, so $7 \mid 2^n - 1 \iff 3 \mid n$, so $21 \mid n$. Now we must ensure that 3 does not divide $2^n - 1$. Since $2^n - 1 \equiv (-1)^n - 1 \pmod{3}$, we have $3 \mid (2^n - 1) \iff n$ is even. Thus, n must be an odd multiple of 21. Finally, since the order of 2 modulo 5 is 4, $5 \mid 2^n - 1 \iff 4 \mid n$. Every multiple of $84 = \text{lcm}(21, 4)$ is even, so no odd multiple of 21 has $5 \mid 2^n - 1$. Therefore, the desired integers are odd multiples of 21 not exceeding 1000. There are $\lfloor \frac{1000}{21} \rfloor = 47$ multiples of 21, of which 24 are odd multiples. Hence, the answer is

$$\boxed{24}$$

proposed by Karam

16. Let $x = \log_a(b)$ and $y = \log_b(c)$. Using the change of base property, we know that $\log_a(c) = \log_a(b) \cdot \log_b(c) = xy$. We can translate the given equation $\log_a(b) + \log_b(c) = \log_a(c) = 2^k$ into a system of equations namely

$$\begin{aligned}x + y &= 2^k \\xy &= 2^k\end{aligned}$$

Because we have the sum and product of x and y , they must be the roots of the quadratic equation

$$t^2 - 2^k t + 2^k = 0$$

Applying the quadratic formula gives

$$t = \frac{2^k \pm \sqrt{2^{2k} - 4(2^k)}}{2} = 2^{k-1} \pm \sqrt{2^{2k-2} - 2^k}$$

We are given that a, b , and c are positive integers plus $a \geq 2$ and $b \geq 2$. Because $x = \log_a(b)$, we can rewrite this as $b = a^x$. Notice that x is a root of a polynomial with integer coefficients ($t^2 - 2^k t + 2^k = 0$), which makes x an algebraic number. By Gelfond-Schneider theorem, if x were an irrational algebraic number, a^x would be a transcendental number however, we are given that b is an integer, meaning b must be algebraic this creates a contradiction and proves that x must be a rational number. For the root x to be rational, the expression inside the square root (the discriminant) must be a perfect square namely

$$2^{2k-2} - 2^k = 2^k(2^{k-2} - 1) = w^2$$

for some integer $w \geq 0$. Let's analyze the possible values for k

- If $k = 1$: The discriminant evaluates to $2^1(2^{-1} - 1) = -1$, yielding complex roots.
- If $k = 2$: The discriminant evaluates to $2^2(2^0 - 1) = 0$, which is a perfect square. This yields a single, rational double-root.
- If $k \geq 3$:

Case A (Even k): Let $k = 2m$ where $m \geq 2$. For $2^{2m}(2^{2m-2} - 1)$ to be a perfect square, the odd term $2^{2m-2} - 1$ must be a perfect square. The only perfect squares that differ by 1 are 0 and 1, meaning $2^{2m-2} - 1 = 0 \implies m = 1 \implies k = 2$. This contradicts $k \geq 3$.

Case B (Odd k): Let $k = 2m + 1$ where $m \geq 1$. The discriminant is $2^{2m+1}(2^{2m-1} - 1)$. For this to be a square, $2(2^{2m-1} - 1)$ must be a square, which requires $2^{2m-1} - 1 = 2r^2$ for some integer r . If $m = 1$, $2^1 - 1 = 1$, which does not equal $2r^2$. If $m \geq 2$, 2^{2m-1} is a multiple of 8, meaning $2^{2m-1} - 1 \equiv -1 \equiv 7 \pmod{8}$. However, $2r^2 \pmod{8}$ can only ever be 0 or 2. Thus, there are no solutions for odd k .

This proves that $k = 2$ is the unique solution that satisfies the conditions. For $k = 2$, the double root of our quadratic is $x = y = 2$. We need to evaluate $\log_{ab}(c) - 1$ we get

$$\log_{ab}(c) = \frac{\log_a(c)}{\log_a(ab)} = \frac{\log_a(b) \cdot \log_b(c)}{\log_a(a) + \log_a(b)} = \frac{xy}{1+x}$$

Substituting $x = 2$ and $y = 2$ yields

$$\log_{ab}(c) = \frac{2 \cdot 2}{1 + 2} = \frac{4}{3}$$

Therefore the expression that we want is

$$\log_{ab}(c) - 1 = \frac{4}{3} - 1 = \frac{1}{3} \implies 1 + 3 = \boxed{4}$$

Proposed by Shining

17. Let the angles of $\triangle ABC$ be α, β , and γ . Because AD is an altitude, $\triangle ABD$ and $\triangle ACD$ are right triangles. We can easily find the side lengths of $\triangle ABC$, in $\triangle ABD$: $AB = \sqrt{AD^2 + BD^2} = \sqrt{24^2 + 10^2} = 26$. In $\triangle ACD$: $AC = \sqrt{AD^2 + CD^2} = \sqrt{24^2 + 18^2} = 30$. Now, we can find the sine and cosine of angles β and γ

- $\sin \beta = \frac{AD}{AB} = \frac{24}{26} = \frac{12}{13}$
- $\cos \beta = \frac{BD}{AB} = \frac{10}{26} = \frac{5}{13}$
- $\sin \gamma = \frac{AD}{AC} = \frac{24}{30} = \frac{4}{5}$
- $\cos \gamma = \frac{CD}{AC} = \frac{18}{30} = \frac{3}{5}$

Since $\alpha = 180^\circ - (\beta + \gamma)$, we can find $\cos \alpha$ to be

$$\begin{aligned}\cos \alpha &= -\cos(\beta + \gamma) = -(\cos \beta \cos \gamma - \sin \beta \sin \gamma) \\ \cos \alpha &= -\left(\frac{5}{13} \cdot \frac{3}{5} - \frac{12}{13} \cdot \frac{4}{5}\right) = -\left(\frac{15}{65} - \frac{48}{65}\right) = \frac{33}{65}\end{aligned}$$

Circle ω has diameter $AD = 24$, so its center O is the midpoint of AD , and its radius is $R = 12$. Because $\angle AED$ and $\angle AFD$ are inscribed angles that subtend the diameter AD , they must be 90° . This means E and F are simply the feet of the altitudes from D to AB and AC . Furthermore, P is the intersection of the tangents to ω at E and F . Namely, P is the pole of chord EF with respect to ω . In $\triangle OEF$, $OE = OF = R = 12$. The line OP is the perpendicular bisector of EF , meaning OP bisects $\angle EOF$. Since PE is tangent to ω we have $\angle OEP = 90^\circ$. First, we find the central angles of ω : $\triangle OAE$ is isosceles ($OA = OE$), so $\angle OAE = \angle BAD = 90^\circ - \beta$. The exterior angle at O is $\angle EOA = 180^\circ - 2(90^\circ - \beta) = 2\beta$. By identical logic on the other side, $\angle FOA = 2\gamma$. The total angle subtended by the major arc EF (the one containing A) is $\angle EOA + \angle FOA = 2\beta + 2\gamma$. The minor arc EF (not containing A) subtends an angle of $360^\circ - (2\beta + 2\gamma) = 2(180^\circ - (\beta + \gamma)) = 2\alpha$. Since OP bisects the minor arc EF , we have $\angle EOP = \alpha$. In the right triangle $\triangle OEP$ we have

$$\begin{aligned}\cos(\angle EOP) &= \frac{OE}{OP} \implies \cos \alpha = \frac{12}{OP} \\ OP &= \frac{12}{\cos \alpha} = \frac{12}{33/65} = \frac{12 \cdot 65}{33} = \frac{260}{11}\end{aligned}$$

To find the distance AP , we will use the Law of Cosines on $\triangle AOP$. We need the angle between the ray OA and the ray OP . Imagine looking from O towards A (let's call this 0°). Ray OE is at an angle of 2β to one side, Ray OF is at an angle of 2γ to the other side. The ray bisecting the major arc EF would be exactly halfway between them, at an angle of $\frac{2\beta + 2\gamma}{2} = \beta + \gamma$ relative to OA . Since OP bisects the minor arc EF , it points in the exact opposite direction of the major arc bisector therefore, the angle $\angle AOP$ is $180^\circ - (\beta + \gamma)$. We now find it's cosine

$$\begin{aligned}\cos(\angle AOP) &= \cos(180^\circ - (\beta + \gamma)) = -\cos(\beta + \gamma) \\ \cos(\beta + \gamma) &= \cos \beta \cos \gamma - \sin \beta \sin \gamma = \left(\frac{5}{13}\right)\left(\frac{3}{5}\right) - \left(\frac{12}{13}\right)\left(\frac{4}{5}\right) = \frac{15}{65} - \frac{48}{65} = -\frac{33}{65} \\ \cos(\angle AOP) &= -\left(-\frac{33}{65}\right) = \frac{33}{65}\end{aligned}$$

Now we apply the Law of Cosines to $\triangle AOP$ to get

$$\begin{aligned}AP^2 &= OA^2 + OP^2 - 2(OA)(OP)\cos(\angle AOP) \\ AP^2 &= 12^2 + \left(\frac{260}{11}\right)^2 - 2(12)\left(\frac{260}{11}\right)\left(\frac{33}{65}\right)\end{aligned}$$

the last term is $2 \cdot 12 \cdot \frac{260}{11} \cdot \frac{33}{65} = 24 \cdot \frac{4}{11} \cdot 63 = \frac{6048}{11}$.

$$AP^2 = 144 + \frac{67600}{121} + \frac{6048}{11}$$

now we add everything up

$$AP^2 = \frac{17424}{121} + \frac{67600}{121} + \frac{66528}{121} = \frac{151552}{121}$$

$$AP = \frac{\sqrt{151552}}{11} = \frac{64\sqrt{37}}{11} \implies a + b + c = 64 + 37 + 11 = \boxed{112}$$

Proposed by Karam

18. We can interpret this algebraic system geometrically using the Law of Cosines. For a triangle with side lengths A and B that meet at an angle of 120° , the length of the opposite side C satisfies

$$C^2 = A^2 + B^2 - 2AB \cos(120^\circ) = A^2 + AB + B^2$$

Thus, the system of equations describes a point P inside an outer triangle $\triangle A_1A_2A_3$ such that the segments connecting P to the three vertices have lengths x , y , and z , and all three central angles meeting at P are exactly 120° . This configuration means P is the Fermat-Torricelli point of the outer triangle. The side lengths of the outer triangle $\triangle A_1A_2A_3$ are given by the square roots namely

- $c = \sqrt{x^2 + xy + y^2} = \sqrt{40}$
- $a = \sqrt{y^2 + yz + z^2} = \sqrt{25} = 5$
- $b = \sqrt{z^2 + zx + x^2} = \sqrt{49} = 7$

The total area of the outer triangle is the sum of the areas of the three internal triangles meeting at P , each having a vertex angle of 120° :

$$\text{Area} = \frac{1}{2}xy \sin(120^\circ) + \frac{1}{2}yz \sin(120^\circ) + \frac{1}{2}zx \sin(120^\circ)$$

$$\text{Area} = \frac{\sqrt{3}}{4}(xy + yz + zx) = \frac{\sqrt{3}}{4}S$$

We rewrite this relation to isolate our target value S

$$S = \frac{4}{\sqrt{3}} \times \text{Area}$$

Next, we calculate the area of the outer triangle with side lengths 5, 7, and $\sqrt{40}$ using Heron's formula. The semiperimeter s is

$$s = \frac{5 + 7 + \sqrt{40}}{2} = 6 + \sqrt{10}$$

Applying Heron's gives

$$\text{Area} = \sqrt{s(s-a)(s-b)(s-c)}$$

$$\text{Area} = \sqrt{(6 + \sqrt{10}) \cdot (6 + \sqrt{10} - 5) \cdot (6 + \sqrt{10} - 7) \cdot (6 + \sqrt{10} - \sqrt{40})}$$

$$\text{Area} = \sqrt{(6 + \sqrt{10})(1 + \sqrt{10})(-1 + \sqrt{10})(6 - \sqrt{10})}$$

Grouping the conjugate pairs gives

$$\text{Area} = \sqrt{\left[(6 + \sqrt{10})(6 - \sqrt{10})\right] \cdot \left[(\sqrt{10} + 1)(\sqrt{10} - 1)\right]}$$

$$\text{Area} = \sqrt{(36 - 10)(10 - 1)} = \sqrt{26 \times 9} = 3\sqrt{26}$$

Finally, we substitute the area back into the expression for S to arrive at

$$S = \frac{4}{\sqrt{3}} \cdot 3\sqrt{26} = 4\sqrt{3} \cdot \sqrt{26} = 4\sqrt{78} \implies a + b = 4 + 78 = \boxed{82}.$$

Proposed by Karam

19. Let the total number of paths of length N that return to the origin be W_N which only works when N is even so let $N = 2n$. Note that we need an equal number of right steps, left steps and an equal number of up steps, down steps so we have

$$W_{2n} = \binom{2n}{n}^2$$

such paths. Let f_{2k} be the number of paths of length $2k$ where the walk returns to the origin at step $2k$ for the first time. If the first return to $(0, 0)$ occurs at step $2k$ (where $2 \leq 2k \leq 2n$) then we have the remainder of the path is simply a path of length $2n - 2k$ starting and ending at the origin. So we arrive at the following

$$W_{2n} = \sum_{k=1}^n f_{2k} W_{2n-2k}$$

see that from this we get

$$f_{2n} = W_{2n} - \sum_{k=1}^{n-1} f_{2k} W_{2n-2k}.$$

Now $W_2 = \binom{2}{1}^2 = 4$ and $W_4 = \binom{4}{2}^2 = 36$ and $W_6 = \binom{6}{3}^2 = 400$ and $W_8 = \binom{8}{4}^2 = 4900$ and $W_{10} = \binom{10}{5}^2 = 63504$. Again these calculations look nasty but taking modulo 1000 makes it way simpler. See that f_2 : $W_2 = 4$ and f_4 : $W_4 - f_2 W_2 = 36 - (4)(4) = 20$ and f_6 : $W_6 - (f_2 W_4 + f_4 W_2) = 400 - (4 \cdot 36 + 20 \cdot 4) = 400 - 224 = 176$ and f_8 : $W_8 - (f_2 W_6 + f_4 W_4 + f_6 W_2) = 4900 - (4 \cdot 400 + 20 \cdot 36 + 176 \cdot 4) = 4900 - 3024 = 1876$ and finally we compute f_{10} :

$$W_{10} - (f_2 W_8 + f_4 W_6 + f_6 W_4 + f_8 W_2) = 63504 - (4 \cdot 4900 + 20 \cdot 400 + 176 \cdot 36 + 1876 \cdot 4) = 63504 - 41440 = 22064$$

taking modulo 1000 gives

$$\boxed{64}.$$

proposed by Karam

20. Let the original number N have k digits. We can separate N into its last digit, d , and the remaining part at the front which we will call x . See that $0 \leq d \leq 9$ and because N is a positive integer and we are moving d to the front to create another integer that is larger ($\frac{13}{4}N$), d cannot be 0. We can express N algebraically as $N = 10x + d$. Since N has k digits, x must have $k - 1$ digits. When we move the rightmost digit d to the very front, the new number N' is formed by shifting d by $k - 1$ place values namely

$$N' = d \cdot 10^{k-1} + x$$

The problem states that $N' = \frac{13}{4}N$. To avoid fractions, we write this as

$$4N' = 13N$$

Substituting yields

$$4(d \cdot 10^{k-1} + x) = 13(10x + d)$$

$$4d \cdot 10^{k-1} + 4x = 130x + 13d$$

Isolate the terms with x on one side and d on the other to get

$$4d \cdot 10^{k-1} - 13d = 126x$$

$$d(4 \cdot 10^{k-1} - 13) = 126x$$

Solving for x , we get

$$x = \frac{d(4 \cdot 10^{k-1} - 13)}{126}$$

For x to be an integer, the numerator $d(4 \cdot 10^{k-1} - 13)$ must be divisible by 126. The prime factorization of 126 is $126 = 2 \cdot 9 \cdot 7$ so we must test the divisibility of our numerator with 2, 9, and 7. We first check modulo

2 (which is just parity). For any $k \geq 2$, 10^{k-1} is even, so $4 \cdot 10^{k-1}$ is even, subtracting 13 makes the term $(4 \cdot 10^{k-1} - 13)$ odd. Because 126 is even, the factor of 2 must come from d thus d must be an even digit

$$d \in \{2, 4, 6, 8\}$$

Let's check the odd term modulo 9. We know $10 \equiv 1 \pmod{9}$, so $10^{k-1} \equiv 1 \pmod{9}$. So we have

$$4(1) - 13 = -9 \equiv 0 \pmod{9}$$

So $(4 \cdot 10^{k-1} - 13)$ is always divisible by 9, regardless of the value of k . We need the remaining factor of 7 to divide $d(4 \cdot 10^{k-1} - 13)$. Since $d \in \{2, 4, 6, 8\}$, d is never divisible by 7. Thus, we must have

$$4 \cdot 10^{k-1} - 13 \equiv 0 \pmod{7}$$

We can rewrite this as

$$4 \cdot 3^{k-1} + 1 \equiv 0 \pmod{7}$$

$$4 \cdot 3^{k-1} \equiv -1 \equiv 6 \pmod{7}$$

Multiply both sides by 2 to get

$$3^{k-1} \equiv 12 \equiv 5 \pmod{7}$$

Now, let's look at the powers of 3 (mod 7) they are

- $3^1 \equiv 3$
- $3^2 \equiv 2$
- $3^3 \equiv 6$
- $3^4 \equiv 4$
- $3^5 \equiv 5 \leftarrow$ this is our target.
- $3^6 \equiv 1$

For $3^{k-1} \equiv 5 \pmod{7}$, the exponent must satisfy $k-1 \equiv 5 \pmod{6}$, which simplifies to

$$k \equiv 0 \pmod{6}$$

The number of digits in N , k , must be a multiple of 6. Let $k = 6m$ for some positive integer m . Let's plug $k = 6m$ back into our equation for N to get

$$N = 10x + d = 10 \left(\frac{d(4 \cdot 10^{6m-1} - 13)}{126} \right) + d$$

$$N = \frac{40d \cdot 10^{6m-1} - 130d + 126d}{126}$$

$$N = \frac{4d \cdot 10^{6m} - 4d}{126} = \frac{2d(10^{6m} - 1)}{63}$$

The original number N must have exactly $k = 6m$ digits. This means $10^{6m-1} \leq N < 10^{6m}$. Let's test our valid even digits $\{2, 4, 6, 8\}$ in our formula for N . If $d = 2$ then

$$N = \frac{4(10^{6m} - 1)}{63} \approx 0.0634 \times 10^{6m}$$

Because $0.0634 < 0.1$, N is strictly less than 10^{6m-1} . This means if $d = 2$, the resulting N will only have $6m - 1$ digits thus $d = 2$ is an extraneous root and is invalid. So the only valid digits are $d \in \{4, 6, 8\}$. Let $U_m = \frac{10^{6m} - 1}{63}$. For $m = 1$, $U_1 = \frac{99999}{63} = 15873$. For $m = 2$, $U_2 = \frac{9999999999}{63} = 15873015873$. In general, U_m is the block '015873' repeated m times (with the leading zero dropped on the very first block). For any given m , our valid integers N are

- $N_{m,1} = 2(4)U_m = 8U_m$
- $N_{m,2} = 2(6)U_m = 12U_m$
- $N_{m,3} = 2(8)U_m = 16U_m$

Because $U_1 < U_2 < U_3 \dots$, the infinite sequence $N_1 < N_2 < N_3 \dots$ is formed by taking $m = 1, 2, 3 \dots$ in groups of three

- $N_1 = 8U_1$
- $N_2 = 12U_1$
- $N_3 = 16U_1$
- $N_4 = 8U_2$
- ...

We are looking for the 25th term in this sequence. Since $25 = 3(8) + 1$, N_{25} corresponds to the first term of the 9th group ($m = 9$). See that

$$N_{25} = 8U_9$$

We need to evaluate $8U_9 \pmod{1000}$. Remember, the structure of U_m is a repeating block of 15873 namely

$$U_9 = 15873015873\dots015873$$

Regardless of how massive m gets, the last six digits of U_m will always be 015873 so the last three digits of U_9 are 873. Thus

$$U_9 \equiv 873 \pmod{1000}$$

$$N_{25} \equiv 8(873) \equiv 6984 \equiv \boxed{984} \pmod{1000}$$

Proposed by Phillip

Solutions to problems 12 and 16 through 20 written by Karam. Solutions to problems 1,3,5,...15 written by Vajpai. Solutions to problems 2,4,6,8,10,14 written by Bryan.

Div 1 Test A

1. Triangle BDC is isosceles. Thus

$$\angle DBC = \angle BCD = 40^\circ.$$

Therefore,

$$\angle ABD = 75^\circ - 40^\circ = \boxed{35^\circ}.$$

proposed by karam.

2. Let $y = 2^x$ we express every term in terms of y .

- $8^x = (2^3)^x = (2^x)^3 = y^3$
- $4^x = (2^2)^x = (2^x)^2 = y^2$
- $4^{x+1} = 4 \cdot 4^x = 4y^2$
- $2^{x+1} = 2 \cdot 2^x = 2y$

Substituting into the original equation gives

$$y^3 + y^2 = 4y^2 + 2y \implies y^3 - 3y^2 - 2y = 0 \implies y(y^2 - 3y - 2) = 0 \implies y^2 - 3y - 2 = 0$$

. We solve for y using quadratic formula and get

$$y = \frac{3 \pm \sqrt{(-3)^2 - 4(1)(-2)}}{2} = \frac{3 \pm \sqrt{17}}{2}$$

we take the positive root as $y = 2^x > 0$ so

$$y = \frac{3 + \sqrt{17}}{2}.$$

Notice that

$$4^{2x} = (2^2)^{2x} = (2^x)^4 = y^4$$

so

$$y^2 = \left(\frac{3 + \sqrt{17}}{2} \right)^2 = \frac{9 + 6\sqrt{17} + 17}{4} = \frac{26 + 6\sqrt{17}}{4} = \frac{13 + 3\sqrt{17}}{2}.$$

We square again to get

$$y^4 = \left(\frac{13 + 3\sqrt{17}}{2} \right)^2 = \frac{169 + 78\sqrt{17} + 9(17)}{4} = \frac{169 + 78\sqrt{17} + 153}{4} = \frac{322 + 78\sqrt{17}}{4}.$$

Next we simplify the fraction to arrive at

$$4^{2x} = \frac{161 + 39\sqrt{17}}{2} \implies a + b + c + d = 161 + 39 + 17 + 2 = \boxed{219}.$$

proposed by shining

3. There are

$$\binom{27}{2} = 351$$

pairs of blocks. A laser can hit both blocks only if they lie in the same row.

There are $3 \cdot 3^2 = 27$ rows, and each row contains $\binom{3}{2} = 3$ pairs. Thus the answer is

$$351 - 27\binom{3}{2} = \boxed{270}.$$

proposed by karam.

4. Let $d(n)$ be the number of positive divisors of n . The condition states

$$d(n) \geq \frac{n}{2}$$

Because divisors pair up, the number of divisors cannot exceed $2\sqrt{n}$ (because every divisor d from $1, 2, \dots, \sqrt{n}$ is paired up with $\frac{n}{d}$ from the set $\sqrt{n} + 1, \dots, n$) so we get

$$d(n) \leq 2\sqrt{n}$$

Combining these two gives

$$\frac{n}{2} \leq 2\sqrt{n} \implies n \leq 4\sqrt{n} \implies n^2 \leq 16n \implies n \leq 16$$

We test all integers $1 \leq n \leq 16$ to see if $d(n) \geq n/2$:

- $n = 1$: Divisors are $\{1\}$. So $d(1) = 1$, which is ≥ 0.5 (Valid)
- $n = 2$: Divisors are $\{1, 2\}$. So $d(2) = 2$, which is ≥ 1 (Valid)
- $n = 3$: Divisors are $\{1, 3\}$. So $d(3) = 2$, which is ≥ 1.5 (Valid)
- $n = 4$: Divisors are $\{1, 2, 4\}$. So $d(4) = 3$, which is ≥ 2 (Valid)
- $n = 5$: Divisors are $\{1, 5\}$. So $d(5) = 2$, which is < 2.5 (Invalid)
- $n = 6$: Divisors are $\{1, 2, 3, 6\}$. So $d(6) = 4$, which is ≥ 3 (Valid)
- $n = 7$: Divisors are $\{1, 7\}$. So $d(7) = 2$, which is < 3.5 (Invalid)
- $n = 8$: Divisors are $\{1, 2, 4, 8\}$. So $d(8) = 4$, which is ≥ 4 (Valid)
- $n = 9$: Divisors are $\{1, 3, 9\}$. So $d(9) = 3$, which is < 4.5 (Invalid)
- $n = 10$: Divisors are $\{1, 2, 5, 10\}$. So $d(10) = 4$, which is < 5 (Invalid)
- $n = 11$: Divisors are $\{1, 11\}$. So $d(11) = 2$, which is < 5.5 (Invalid)
- $n = 12$: Divisors are $\{1, 2, 3, 4, 6, 12\}$. So $d(12) = 6$, which is ≥ 6 (Valid)
- $n = 13$: Divisors are $\{1, 13\}$. So $d(13) = 2$, which is < 6.5 (Invalid)
- $n = 14$: Divisors are $\{1, 2, 7, 14\}$. So $d(14) = 4$, which is < 7 (Invalid)
- $n = 15$: Divisors are $\{1, 3, 5, 15\}$. So $d(15) = 4$, which is < 7.5 (Invalid)
- $n = 16$: Divisors are $\{1, 2, 4, 8, 16\}$. So $d(16) = 5$, which is < 8 (Invalid)

The valid n values are 1, 2, 3, 4, 6, 8, and 12. Summing these values together gives

$$1 + 2 + 3 + 4 + 6 + 8 + 12 = \boxed{36}$$

proposed by val.

5. One of the ten dice is a fair 10sided die. Fix the results of the other nine dice, and let their sum be S . We need the 10sided die to show a number r such that

$$S + r \equiv 1 \pmod{10}.$$

The possible rolls $1, 2, \dots, 10$ represent every residue modulo 10 exactly once. Regardless of the value of S , exactly one of the ten possible rolls makes the total have units digit 1.

Hence

$$P(\text{units digit } 1) = \frac{1}{10}.$$

So the answer is

$$m + n = 1 + 10 = \boxed{11}.$$

proposed by Rayoon

6. Let x be a positive integer, we want to find the number of perfect cubes $x^3 < 10^6$ that cannot be written in the form y^2 for any integer y . First, we find the total number of positive perfect cubes less than 10^6

$$x^3 < 10^6 \implies x < (10^6)^{1/3} \implies x < 100$$

Since x is a positive integer, x can range from 1 to 99 giving us a total of 99 perfect cubes. Next, a number is both a perfect cube and a perfect square if and only if it is a perfect sixth power m^6 . We count how many of our 99 perfect cubes are also perfect squares. This happens when the base x itself is a perfect square $x = m^2$ which means

$$m^2 < 100 \implies m < 10$$

The possible positive integer values for m are 1, 2, 3, 4, 5, 6, 7, 8, and 9 thus there are exactly 9 perfect cubes that are also perfect squares. To find the perfect cubes that are not perfect squares, we subtract the overlapping sixth powers from the total number of cubes to arrive at our final answer

$$\text{Total Cubes} - \text{Cubes that are Squares} = 99 - 9 = \boxed{90}$$

proposed by Shining+Karam

7. Let

$$N = 2026^{2026}.$$

Since

$$2026 = 2 \cdot 1013$$

$$N = 2^{2026} \cdot 1013^{2026}.$$

Every positive divisor of N has the form

$$2^a 1013^b, \quad 0 \leq a, b \leq 2026.$$

The total number of positive divisors is

$$(2026 + 1)^2 = 2027^2 = 4,108,729.$$

$$\sqrt{N} = 2026^{1013}.$$

Every divisor $d \neq \sqrt{N}$ can be paired with N/d . Exactly one divisor in each pair is less than $\sqrt{N}$. Since $\sqrt{N}$ is the only unpaired divisor, the number of divisors less than $\sqrt{N}$ is

$$\frac{4,108,729 - 1}{2} = 2,054,364.$$

Desired probability is

$$\frac{2,054,364}{4,108,729}.$$

So only adding the last 3 digits,

$$m + n = 364 + 729 = 1093$$

The last three digits are

$$\boxed{093}.$$

proposed by Neil

8. Let N be the maximum number of players, and let the challenges be numbered $1, 2, \dots, 8$, each player chooses a subset of 3 distinct challenges. First, we determine the maximum number of times any single challenge can be chosen across all players. Suppose a specific challenge is chosen by k players. Because no two players can share a second challenge in common, the other 2 challenges chosen by each of these k players must all be distinct. The total number of remaining challenges required by these k players is $2k$. Since there are only 7 other challenges available in the entire pool, we must have

$$2k \leq 7 \implies k \leq 3.5$$

Since k must be an integer, any single challenge can appear in at most 3 players' selections. Next, we bound the total number of players. Since there are 8 challenges and each can be used a maximum of 3 times, there are at most $8 \times 3 = 24$ available slots in total. Since each player must take exactly 3 challenges, the maximum number of players is

$$N \leq \frac{24}{3} = 8$$

To prove that 8 players is achievable, we construct a valid arrangement.

- Player 1: $\{8, 1, 2\}$
- Player 2: $\{8, 3, 4\}$
- Player 3: $\{8, 5, 6\}$
- Player 4: $\{7, 1, 3\}$
- Player 5: $\{7, 2, 5\}$
- Player 6: $\{7, 4, 6\}$
- Player 7: $\{1, 4, 5\}$
- Player 8: $\{2, 3, 6\}$

Checking the intersections confirms that no two players share more than 1 challenge, thus, the maximum number of players is $\boxed{8}$.

proposed by Val

9. Let

$$d = \gcd(m, n), \quad m = da, \quad n = db,$$

where $\gcd(a, b) = 1$. Then

$$\text{lcm}(m, n) = dab.$$

Substituting into the given equation gives

$$dab = da + db + d + 6,$$

so

$$d(ab - a - b - 1) = 6.$$

Set

$$e = ab - a - b - 1.$$

Since $de = 6$ and we are dealing with integers,

$$e \in \{1, 2, 3, 6\} \quad \text{and} \quad d = \frac{6}{e}.$$

The equation for a and b can be rewritten as

$$(a - 1)(b - 1) = e + 2.$$

We now consider the four possibilities. Casework: If $e = 1$, then

$$(a - 1)(b - 1) = 3.$$

This gives $(a, b) = (2, 4)$ or $(4, 2)$, neither of which is coprime (we are looking for coprime numbers a and b because we assume d is the only common divisors and a or b does not divide the other).

If $e = 2$, then

$$(a - 1)(b - 1) = 4.$$

The coprime possibilities are

$$(a, b) = (2, 5), (5, 2).$$

Since $d = 3$, these give

$$(m, n) = (6, 15), (15, 6).$$

If $e = 3$, then

$$(a - 1)(b - 1) = 5.$$

This gives $(a, b) = (2, 6)$ or $(6, 2)$, neither of which is coprime.

If $e = 6$, then

$$(a - 1)(b - 1) = 8.$$

The coprime possibilities are

$$(a, b) = (2, 9), (3, 5), (5, 3), (9, 2).$$

Since $d = 1$, these give the same four ordered pairs for (m, n) .

Therefore, the six ordered pairs are

$$(6, 15), (15, 6), (2, 9), (3, 5), (5, 3), (9, 2).$$

All satisfy the given bounds, so the answer is

$$\boxed{6}.$$

proposed by Karam

10. We can rewrite the floor function term as

$$\left\lfloor \frac{i^2}{7} \right\rfloor = \frac{i^2 - (i^2 \bmod 7)}{7}$$

Splitting the original summation into two components gives

$$S = \frac{1}{7} \sum_{i=1}^{2026} i^2 - \frac{1}{7} \sum_{i=1}^{2026} (i^2 \bmod 7)$$

Let the first part be $T_1 = \frac{1}{7} \sum_{i=1}^{2026} i^2$ using the formula for the sum of squares yields

$$T_1 = \frac{1}{7} \cdot \frac{2026(2027)(4053)}{6} = \frac{2026 \times 2027 \times 579}{6}$$

$$T_1 = 1013 \times 2027 \times 193$$

Since we only need the final 3 digits, we evaluate $T_1 \pmod{1000}$

$$T_1 \equiv 13 \times 27 \times 193 \pmod{1000}$$

$$T_1 \equiv 351 \times 193 = 67743 \equiv 743 \pmod{1000}$$

Let the second part be $T_2 = \frac{1}{7} \sum_{i=1}^{2026} (i^2 \pmod{7})$. The remainders of $i^2 \pmod{7}$ repeat in a period of 7

$$\{1, 4, 2, 2, 4, 1, 0\}$$

The sum of a single full period is $1 + 4 + 2 + 2 + 4 + 1 + 0 = 14$. Dividing the total by the period length 7 gives

$$2026 = 289 \times 7 + 3$$

The sum consists of 289 complete periods and the first 3 terms of the next period (1, 4, 2) which sums to

$$\sum_{i=1}^{2026} (i^2 \pmod{7}) = (289 \times 14) + (1 + 4 + 2) = 4046 + 7 = 4053$$

$$T_2 = \frac{4053}{7} = 579$$

Subtracting T_2 from T_1 to determine the final value modulo 1000 gives

$$S \equiv 743 - 579 = \boxed{164} \pmod{1000}$$

proposed by Shining

11. See $\triangle ABC$ is a right-angled triangle with $\angle B = 90^\circ$. The inradius r of the right triangle is

$$r = \frac{BC + AB - AC}{2} = \frac{3 + 4 - 5}{2} = 1$$

Thus, the perpendicular distance from I to both sides AB and BC is exactly 1.

By the Angle Bisector Theorem:

1. For point D on AB (bisector CD):

$$\frac{AD}{DB} = \frac{AC}{BC} = \frac{5}{3} \implies DB = 4 \times \frac{3}{5+3} = \frac{3}{2}$$

2. For point E on BC (bisector AE):

$$\frac{CE}{EB} = \frac{AC}{AB} = \frac{5}{4} \implies EB = 3 \times \frac{4}{5+4} = \frac{4}{3}$$

Using the Pythagorean theorem on the right triangle $\triangle DBE$ gives

$$DE = \sqrt{BD^2 + BE^2} = \sqrt{\left(\frac{3}{2}\right)^2 + \left(\frac{4}{3}\right)^2} = \sqrt{\frac{9}{4} + \frac{16}{9}} = \sqrt{\frac{145}{36}} = \frac{\sqrt{145}}{6}$$

The distance d from I to DE is the height of $\triangle IDE$ from vertex I . The area of $\triangle IDE$ is

$$\text{Area}(\triangle IDE) = \frac{1}{2} \times DE \times d$$

Using the areas of the surrounding triangles

$$\begin{aligned}\text{Area}(\triangle DBE) &= \frac{1}{2} \times BD \times BE = \frac{1}{2} \times \frac{3}{2} \times \frac{4}{3} = 1 \\ \text{Area}(\triangle IBD) &= \frac{1}{2} \times BD \times r = \frac{1}{2} \times \frac{3}{2} \times 1 = \frac{3}{4} \\ \text{Area}(\triangle IBE) &= \frac{1}{2} \times BE \times r = \frac{1}{2} \times \frac{4}{3} \times 1 = \frac{2}{3}\end{aligned}$$

Since I is further from the vertex B than the segments BD and BE , the area relationship is

$$\text{Area}(\triangle IDE) = \text{Area}(\triangle IBD) + \text{Area}(\triangle IBE) - \text{Area}(\triangle DBE)$$

$$\text{Area}(\triangle IDE) = \frac{3}{4} + \frac{2}{3} - 1 = \frac{9 + 8 - 12}{12} = \frac{5}{12}$$

Equating the two expressions for $\text{Area}(\triangle IDE)$ gives

$$\begin{aligned}\frac{5}{12} &= \frac{1}{2} \times \frac{\sqrt{145}}{6} \times d \\ \frac{5}{12} &= \frac{\sqrt{145}}{12} \times d \implies d = \frac{5}{\sqrt{145}} = \frac{\sqrt{145}}{29} \implies 145 + 29 = \boxed{174}\end{aligned}$$

Proposed by Rayoon

12. Let $N = \frac{2026!}{11^a \cdot 13^b}$. Since $143 = 11 \times 13$, we can solve for $N \pmod{11}$ and $N \pmod{13}$ separately, then combine them using the Chinese Remainder Theorem. Using Legendre's Formula we can compute the exponents,

$$\begin{aligned}a &= \left\lfloor \frac{2026}{11} \right\rfloor + \left\lfloor \frac{2026}{121} \right\rfloor + \left\lfloor \frac{2026}{1331} \right\rfloor = 184 + 16 + 1 = 201 \\ b &= \left\lfloor \frac{2026}{13} \right\rfloor + \left\lfloor \frac{2026}{169} \right\rfloor = 155 + 11 = 166\end{aligned}$$

We isolate the prime 11 component from the rest of the expression:

$$N = \frac{2026!}{11^{201}} \times \frac{1}{13^{166}} \pmod{11}$$

First, we evaluate the factorial part using the generalization of Wilson's Theorem, writing 2026 in base 11 gives digits 1, 5, 8, 2 so

$$\frac{2026!}{11^{201}} \equiv (-1)^{201} \times 1! \times 5! \times 8! \times 2! \pmod{11}$$

- $(-1)^{201} = -1$
- $5! = 120 \equiv -1 \pmod{11}$
- $8! \equiv 5 \pmod{11}$
- $2! = 2$

$$\frac{2026!}{11^{201}} \equiv (-1) \times 1 \times (-1) \times 5 \times 2 = 10 \pmod{11}$$

Second, we evaluate the division by 13^{166} . Since $13 \equiv 2 \pmod{11}$, by Fermat's Little Theorem we have

$$2^{166} \equiv 2^6 = 64 \equiv 9 \pmod{11}$$

The modular inverse of $9 \pmod{11}$ is 5 so

$$N \equiv 10 \times 5 = 50 \equiv 6 \pmod{11}$$

We rewrite the expression isolating the prime 13 component to get

$$N = \frac{2026!}{13^{166}} \times \frac{1}{11^{201}} \pmod{13}$$

Converting 2026 to base 13 gives digits 11, 12, 11 so

$$\frac{2026!}{13^{166}} \equiv (-1)^{166} \times 11! \times 12! \times 11! \pmod{13}$$

- $(-1)^{166} = 1$
- $12! \equiv -1 \pmod{13}$
- $11! \equiv 1 \pmod{13}$

$$\frac{2026!}{13^{166}} \equiv 1 \times 1 \times (-1) \times 1 = -1 \equiv 12 \pmod{13}$$

Next, we evaluate the division by 11^{201} , since $11 \equiv -2 \pmod{13}$ we have

$$(-2)^{201} = -2^{201} \equiv -2^9 = -512 \equiv -5 \equiv 8 \pmod{13}$$

The modular inverse of $8 \pmod{13}$ is 5 so

$$N \equiv 12 \times 5 = 60 \equiv 8 \pmod{13}$$

We now have a system of congruences

- $N \equiv 6 \pmod{11}$
- $N \equiv 8 \pmod{13}$

From the second congruence, $N = 13k + 8$ so

$$13k + 8 \equiv 6 \pmod{11}$$

$$2k \equiv -2 \equiv 9 \equiv 20 \pmod{11} \implies k \equiv 10 \pmod{11}$$

Substituting $k = 10$ back into the formula for N gives us our final answer of

$$N = 13(10) + 8 = \boxed{138}.$$

Proposed by Abhi

13. Recall that $\coth(2\theta) = \frac{\coth^2 \theta + 1}{2 \coth \theta}$ so applying this twice gives

$$\coth(4\theta) = \frac{\coth^2(2\theta) + 1}{2 \coth(2\theta)} = \frac{(\frac{\coth^2 \theta + 1}{2 \coth \theta})^2 + 1}{2(\frac{\coth^2 \theta + 1}{2 \coth \theta})} = \frac{\coth^4 \theta + 6 \coth^2 \theta + 1}{4 \coth^3 \theta + 4 \coth \theta}$$

thus setting $a_n = \coth(\theta_n)$ our recurrence becomes $\coth(\theta_{n+1}) = \coth(4\theta_n)$ so $\theta_{n+1} = 4\theta_n$ thus $\theta_n = 4^{n-1}\theta_1$. We have $\coth(\theta_1) = 2$ so

$$\frac{e^{\theta_1} + e^{-\theta_1}}{e^{\theta_1} - e^{-\theta_1}} = 2 \implies e^{\theta_1} + e^{-\theta_1} = 2e^{\theta_1} - 2e^{-\theta_1} \implies e^{2\theta_1} = 3$$

let $x_n = e^{2\theta_n}$ then $x_1 = 3$ and since $x_{n+1} = e^{2\theta_{n+1}} = e^{2(4\theta_n)} = (e^{2\theta_n})^4 = x_n^4$ so we have $x_n = 3^{4^{n-1}}$. Notice that

$$a_n = \frac{e^{\theta_n} + e^{-\theta_n}}{e^{\theta_n} - e^{-\theta_n}} = \frac{e^{2\theta_n} + 1}{e^{2\theta_n} - 1} = \frac{x_n + 1}{x_n - 1}.$$

Each term in the product is

$$T_n = \frac{4a_n(a_n^2 + 1)}{(a_n + 1)^3}$$

substituting our value for a_n we get

$$a_n + 1 = \frac{2x_n}{x_n - 1}, \quad a_n^2 + 1 = \frac{2(x_n^2 + 1)}{(x_n - 1)^2}$$

so

$$T_n = \frac{4\left(\frac{x_n+1}{x_n-1}\right)\left(\frac{2(x_n^2+1)}{(x_n-1)^2}\right)}{\left(\frac{2x_n}{x_n-1}\right)^3} = \frac{8(x_n+1)(x_n^2+1)}{(x_n-1)^3} \cdot \frac{(x_n-1)^3}{8x_n^3} = \frac{(x_n+1)(x_n^2+1)}{x_n^3} = \left(1 + \frac{1}{x_n}\right) \left(1 + \frac{1}{x_n^2}\right).$$

Since $x_{n+1} = x_n^4$, the sequence of exponents x_n and x_n^2 covers all powers of 2 for the base $y = \frac{1}{x_1} = \frac{1}{3}$ namely

$$P = \left(1 + \frac{1}{3}\right) \left(1 + \frac{1}{9}\right) \left(1 + \frac{1}{81}\right) \left(1 + \frac{1}{6561}\right) \cdots = \prod_{k=0}^{\infty} \left(1 + y^{2^k}\right)$$

so using the identity $\prod_{k=0}^{\infty} (1 + y^{2^k}) = \frac{1}{1-y}$ for $|y| < 1$ we get

$$P = \frac{1}{1 - 1/3} = \frac{1}{2/3} = \frac{3}{2} \implies 3 + 2 = \boxed{5}$$

Proposed by Karam

14. Let $u = x + y$ and $v = x - y$. In each of the 8 steps, Neil moves in one of four directions: $(1, 0)$, $(-1, 0)$, $(0, 1)$, or $(0, -1)$. Let's analyze how each step affects u and v :

- Move $(1, 0) \implies \Delta u = 1, \Delta v = 1$
- Move $(-1, 0) \implies \Delta u = -1, \Delta v = -1$
- Move $(0, 1) \implies \Delta u = 1, \Delta v = -1$
- Move $(0, -1) \implies \Delta u = -1, \Delta v = 1$

Notice that for every possible move, $\Delta u \in \{1, -1\}$ and $\Delta v \in \{1, -1\}$ independently, each with a probability of $\frac{1}{2}$. Thus, u and v are independent, identical 1D random walks of $n = 8$ steps. Solving for x and y gives:

$$x = \frac{u+v}{2}, \quad y = \frac{u-v}{2}$$

Substituting these into the target expression gives

$$x^4 y^2 = \left(\frac{u+v}{2}\right)^4 \left(\frac{u-v}{2}\right)^2 = \frac{1}{64} (u+v)^4 (u-v)^2$$

Expanding the polynomial $(u + v)^4(u - v)^2$ gives

$$(u + v)^4(u - v)^2 = u^6 + 2u^5v - u^4v^2 - 4u^3v^3 - u^2v^4 + 2uv^5 + v^6$$

Since u and v are independent and symmetric about 0, the expected value of any odd power of u or v is zero namely

$$\mathbb{E}[u^5v] = \mathbb{E}[u^3v^3] = \mathbb{E}[uv^5] = 0$$

By symmetry, we also know that $\mathbb{E}[u^6] = \mathbb{E}[v^6]$ and $\mathbb{E}[u^4v^2] = \mathbb{E}[u^2v^4]$. This simplifies our expression to

$$\mathbb{E}[(u + v)^4(u - v)^2] = 2\mathbb{E}[u^6] - 2\mathbb{E}[u^4]\mathbb{E}[u^2]$$

Let $u = \sum_{i=1}^8 X_i$, where $X_i = \pm 1$ with equal probability. Using the standard moment formulas for a 1D random walk with $n = 8$ we get

$$\mathbb{E}[u^2] = n = 8$$

$$\mathbb{E}[u^4] = 3n^2 - 2n = 3(64) - 2(8) = 176$$

$$\mathbb{E}[u^6] = 15n^3 - 30n^2 + 16n = 15(512) - 30(64) + 16(8) = 5888$$

Substitute the computed moments back to get

$$\begin{aligned}\mathbb{E}[(u + v)^4(u - v)^2] &= 2(5888) - 2(176)(8) \\ &= 11776 - 2816 \\ &= 8960\end{aligned}$$

Finally, divide by the factor 64 to get

$$\mathbb{E}[x^4y^2] = \frac{8960}{64} = \boxed{140}$$

Proposed by Karam.

15. Suppose the roster contains k knights and k rabbits. The players can be selected in

$$\binom{9}{k} \binom{11}{k}$$

ways.

After the players are chosen, there are k choices for the knight captain and k choices for the rabbit captain. Thus, for a fixed value of k , the number of rosters is

$$k^2 \binom{9}{k} \binom{11}{k}.$$

Since there are only 9 knights,

$$N = \sum_{k=1}^9 k^2 \binom{9}{k} \binom{11}{k}.$$

Using the absorption identity

$$k \binom{9}{k} = 9 \binom{8}{k-1} \quad \text{and} \quad k \binom{11}{k} = 11 \binom{10}{k-1},$$

we obtain

$$N = 99 \sum_{k=1}^9 \binom{8}{k-1} \binom{10}{k-1}.$$

Let $j = k - 1$. Then

$$N = 99 \sum_{j=0}^8 \binom{8}{j} \binom{10}{j}.$$

Since

$$\binom{10}{j} = \binom{10}{10-j},$$

Vandermonde's identity gives

$$\sum_{j=0}^8 \binom{8}{j} \binom{10}{10-j} = \binom{18}{10} = \binom{18}{8}.$$

$$N = 99 \binom{18}{8} = 99(43758) = 4,332,042.$$

The last three digits are

$$\boxed{042}.$$

Proposed by Abhi

16. We expand both base representations to convert them into base 10

$$N = 1 \cdot b^2 + 2 \cdot b + 3 = b^2 + 2b + 3$$

$$N = 3 \cdot c^2 + 2 \cdot c + 1 = 3c^2 + 2c + 1$$

Equating the two gives

$$b^2 + 2b + 3 = 3c^2 + 2c + 1 \implies b^2 + 2b + 2 = 3c^2 + 2c$$

Completing the square on the LHS yields

$$(b+1)^2 + 1 = 3c^2 + 2c$$

Multiply the equation by 3 and complete the square to get

$$3(b+1)^2 + 3 = 9c^2 + 6c$$

$$3(b+1)^2 + 4 = (3c+1)^2$$

Rearranging this into a standard Pell-like form gives

$$(3c+1)^2 - 3(b+1)^2 = 4$$

Let $X = 3c + 1$ and $Y = b + 1$. The equation simplifies to

$$X^2 - 3Y^2 = 4$$

Since the digits 3 and 2 are used in the representations, the bases must satisfy $b > 3$ and $c > 3$. The positive integer solutions (X_n, Y_n) can be generated from the solution $(2, 0)$ via the linear recurrences

$$X_{n+1} = 2X_n + 3Y_n$$

$$Y_{n+1} = X_n + 2Y_n$$

We now simply check the solutions

- $(X_0, Y_0) = (2, 0) \implies b = -1, c = 1/3$ (Invalid)
- $(X_1, Y_1) = (4, 2) \implies b = 1, c = 1$ (Invalid)

- $(X_2, Y_2) = (14, 8) \implies 3c + 1 = 14 \implies c = 13/3$ (Non-integer)
- $(X_3, Y_3) = (52, 30) \implies 3c + 1 = 52 \implies c = 17$ and $b + 1 = 30 \implies b = 29$ (Valid)
- $(X_4, Y_4) = (194, 112) \implies 3c + 1 = 194 \implies c = 193/3$ (Non-integer)
- $(X_5, Y_5) = (724, 418) \implies 3c + 1 = 724 \implies c = 241$ and $b + 1 = 418 \implies b = 417$ (Valid)
- $(X_6, Y_6) = (2702, 1560) \implies 3c + 1 = 2702 \implies c = 2701/3$ (Non-integer)
- $(X_7, Y_7) = (10084, 5822) \implies b + 1 = 5822 \implies b = 5821$ (Invalid, exceeds 1000)

The valid values for b less than 1000 are 29 and 417 which gives

$$\text{Sum} = 29 + 417 = \boxed{446}$$

Proposed by Karam

17. Using Heron's we derive that the area is 84 and that the circumradius is

$$R = \frac{abc}{4K} = \frac{13 \cdot 14 \cdot 15}{4 \cdot 84} = \frac{65}{8}.$$

Now

$$\cos A = \frac{b^2 + c^2 - a^2}{2bc} = \frac{14^2 + 15^2 - 13^2}{2 \cdot 14 \cdot 15} = \frac{196 + 225 - 169}{420} = \frac{252}{420} = \frac{3}{5}$$

so $\sin A = \sqrt{1 - \left(\frac{3}{5}\right)^2} = \frac{4}{5}$ similarly we arrive at

$$\cos B = \frac{a^2 + c^2 - b^2}{2ac} = \frac{13^2 + 15^2 - 14^2}{2 \cdot 13 \cdot 15} = \frac{169 + 225 - 196}{390} = \frac{198}{390} = \frac{33}{65}$$

$$\sin B = \sqrt{1 - \left(\frac{33}{65}\right)^2} = \frac{56}{65}$$

and

$$\cos C = \frac{a^2 + b^2 - c^2}{2ab} = \frac{13^2 + 14^2 - 15^2}{2 \cdot 13 \cdot 14} = \frac{169 + 196 - 225}{364} = \frac{140}{364} = \frac{5}{13}$$

$$\sin C = \sqrt{1 - \left(\frac{5}{13}\right)^2} = \frac{12}{13}.$$

It is known that $AH = 2R \cos A$ so

$$AH = 2 \left(\frac{65}{8}\right) \left(\frac{3}{5}\right) = \frac{39}{4}$$

now let O be the circumcenter of $\triangle ABC$. It is well known that altitude AD and the circumradius AO are isogonal respect to $\angle A$. So

$$\angle OAH = |\angle B - \angle C|$$

see that $\cos(\angle OAH) = \cos(B - C)$ so

$$\cos(B - C) = \cos B \cos C + \sin B \sin C$$

which is

$$\cos(B - C) = \left(\frac{33}{65}\right) \left(\frac{5}{13}\right) + \left(\frac{56}{65}\right) \left(\frac{12}{13}\right)$$

which simplifies to

$$\cos(B - C) = \frac{165}{845} + \frac{672}{845} = \frac{837}{845}.$$

The center of ω denote it by O_ω must lie on the line containing the circumradius AO . So see that O_ω lies on the perpendicular bisector of the segment AH since it passes through A, H . Let M be the midpoint of AH then consider the right triangle $\triangle AMO_\omega$ so using trig in this triangle we get (see that $\angle MAO_\omega = \angle OAH$) so

$$\cos(\angle OAH) = \frac{AM}{AO_\omega} = \frac{AH/2}{r_\omega}$$

which means

$$2r_\omega = \frac{AH}{\cos(B - C)}$$

thus

$$2r_\omega = \frac{39/4}{837/845} = \frac{39 \cdot 845}{4 \cdot 837}$$

thus we get

$$2r_\omega = \frac{13 \cdot 845}{4 \cdot 279} = \frac{10985}{1116}.$$

By Extended Law of Sines we have

$$PQ = 2r_\omega \sin A$$

so

$$PQ = \left(\frac{10985}{1116} \right) \left(\frac{4}{5} \right)$$

which simplifies to

$$PQ = \frac{2197}{279} \implies m + n = 2197 + 279 = 2 \boxed{476}.$$

Proposed by Karam

18. Let T_n denote the total sum of $S(A)$ over all non-empty subsets of $\{1, 2, 3, \dots, n\}$. We can partition the non-empty subsets of $\{1, 2, 3, \dots, n\}$ into three mutually exclusive categories based on the inclusion of the largest element n .

1. **Subsets not containing n :** These are exactly the non-empty subsets of $\{1, 2, \dots, n-1\}$. The sum of $S(A)$ for these subsets is T_{n-1} .
2. **The single subset $\{n\}$:** For this subset, $S(\{n\}) = n$.
3. **Subsets containing n and at least one element from $\{1, 2, \dots, n-1\}$:** Any such subset can be uniquely represented as $A = A' \cup \{n\}$, where A' is a non-empty subset of $\{1, 2, \dots, n-1\}$. By the definition of $S(A)$, since n is strictly greater than all elements in A' , we have:

$$S(A) = n - 2S(A')$$

There are $2^{n-1} - 1$ possible non-empty subsets A' . Summing over all choices in the third category yields

$$\sum_{A'} (n - 2S(A')) = n(2^{n-1} - 1) - 2T_{n-1}$$

Combining all three categories gives the total sum T_n

$$T_n = T_{n-1} + n + [n(2^{n-1} - 1) - 2T_{n-1}]$$

Simplifying gives

$$T_n = n2^{n-1} - T_{n-1}$$

Solving this recurrence relation with the initial condition $T_1 = 1$, we find the general explicit formula for T_n (one can indeed verify this works, and in contest one can just bash out until the 10th term which isn't that much computation)

$$T_n = \frac{(3n+1)2^n - (-1)^n}{9}$$

Substituting $n = 10$ into this closed form gives

$$T_{10} = \frac{(3 \cdot 10 + 1)2^{10} - (-1)^{10}}{9} = \frac{31 \cdot 1024 - 1}{9} = \frac{31743}{9} = 3527$$

The total number of non-empty subsets of a 10-element set is $2^{10} - 1 = 1023$ so the expected value is

$$E = \frac{T_{10}}{1023} = \frac{3527}{1023} \implies m + n = 3527 + 1023 = 4 \boxed{550}$$

Proposed by Karam

19. We are given the relation

$$P(x^2) = P(x)P(-x)$$

Let r be a complex root of $P(x)$. If $P(r) = 0$, substituting $x = r$ yields $P(r^2) = P(r)P(-r) = 0$. This implies that if r is a root, then r^2 is also a root. Repeatedly applying this logic, the sequence $r, r^2, r^4, r^8, \dots$ must all be roots of $P(x)$. Since a nonconstant polynomial can only have a finite number of roots, this sequence must eventually repeat. This is only possible if $r = 0$ or if $|r| = 1$ (meaning r is a root of unity). Since the multiset of roots of $P(x)$ is perfectly invariant under the mapping $r \mapsto r^2$, the non-zero roots must be roots of unity. We are given that $P(1) \neq 0$, meaning 1 is strictly excluded from being a root. If $P(x)$ had a root of unity of even order, repeatedly squaring it would eventually result in -1 . Squaring -1 then yields 1. Since 1 cannot be a root, $P(x)$ cannot contain any roots of unity of even order. Thus, all roots of $P(x)$ must either be 0 or roots of unity of odd order $k > 1$. Because $P(x)$ has integer coefficients, its roots must be grouped into irreducible polynomials over the rationals. For roots of unity, these are exactly the cyclotomic polynomials $\Phi_k(x)$. Therefore, $P(x)$ must be of the form:

$$P(x) = cx^m \prod_{k \text{ odd}, k > 1} \Phi_k(x)^{e_k}$$

Let's check the leading coefficient c . Let the degree of $P(x)$ be d . The leading term on the left side of $P(x^2) = P(x)P(-x)$ is cx^{2d} , and on the right side, it is $(cx^d) \cdot c(-x)^d = c^2(-1)^d x^{2d}$. Equating these gives $c = c^2(-1)^d$, so $c = (-1)^d$. Because the degree of any cyclotomic polynomial $\Phi_k(x)$ for $k > 1$ is $\phi(k)$ (which is always even), the total degree $d = m + \sum e_k \phi(k)$ has the same parity as m thus, $c = (-1)^m$. This gives us the exact form for any valid polynomial:

$$P(x) = (-1)^m x^m \prod_{k \text{ odd}, k > 1} \Phi_k(x)^{e_k}$$

We are given $-2026 \leq P(2) \leq 2026$. Substituting $x = 2$ into our general form gives

$$P(2) = (-1)^m 2^m \prod \Phi_k(2)^{e_k}$$

Taking the absolute value, we get

$$2^m \prod \Phi_k(2)^{e_k} \leq 2026$$

Let's evaluate $\Phi_k(2)$ for odd $k > 1$:

- $\Phi_3(2) = 2^2 + 2 + 1 = 7$
- $\Phi_5(2) = 2^4 + 2^3 + 2^2 + 2 + 1 = 31$
- $\Phi_7(2) = 2^6 + \dots + 1 = 127$
- $\Phi_9(2) = 2^6 + 2^3 + 1 = 73$
- $\Phi_{11}(2) = 2^{10} + \dots + 1 = 2047$ (Too large, > 2026)
- $\Phi_{15}(2) = \frac{(2^{15}-1)(2-1)}{(2^5-1)(2^3-1)} = \frac{32767}{31 \times 7} = 151$

Any higher odd k results in $\Phi_k(2) > 2026$. So, the only possible cyclotomic factors evaluate to 7, 31, 73, 127, or 151 at $x = 2$. We want to find all combinations of non-negative integers $(m, e_3, e_5, e_7, e_9, e_{15})$ such that the product $V = 2^m \cdot 7^{e_3} \cdot 31^{e_5} \cdot 73^{e_9} \cdot 127^{e_7} \cdot 151^{e_{15}} \leq 2026$. Note that $P(x)$ is nonconstant, so we must exclude the case where $m = 0$ and all $e_k = 0$. We break this into mutually exclusive cases based on the largest prime-related factor present, after extensive computation we arrive at

$$5 + 6 + 7 + 14 + 18 + 10 = \boxed{60}$$

Proposed by Karam

20. We aim to first compute the inner sum then computing the outer sum isn't that bad so we want to find

$$S_n = \sum_{k=1}^n \frac{1}{1 + 2 \cos\left(\frac{2k\pi}{n}\right)}.$$

Let $\omega = e^{2\pi i/n}$ so substituting this into our sum gives

$$\frac{1}{1 + 2 \left(\frac{\omega^k + \omega^{-k}}{2} \right)} = \frac{1}{1 + \omega^k + \omega^{-k}}$$

which simplifies to

$$\frac{\omega^k}{\omega^{2k} + \omega^k + 1}$$

so we want to compute

$$S_n = \sum_{k=1}^n \frac{\omega^k}{\omega^{2k} + \omega^k + 1}.$$

Let $f(z) = \frac{z}{z^2 + z + 1}$ then the roots of the denominator $z^2 + z + 1 = 0$ are

$$\rho = e^{2\pi i/3} = -\frac{1}{2} + i\frac{\sqrt{3}}{2}$$

and

$$\bar{\rho} = e^{-2\pi i/3} = -\frac{1}{2} - i\frac{\sqrt{3}}{2}$$

now we can use partial fraction decomposition. See that we can express this as

$$f(z) = \frac{1}{\rho - \bar{\rho}} \left(\frac{\rho}{z - \rho} - \frac{\bar{\rho}}{z - \bar{\rho}} \right).$$

Apply this to our sum to get

$$S_n = \frac{1}{\rho - \bar{\rho}} \sum_{k=1}^n \left(\frac{\rho}{\omega^k - \rho} - \frac{\bar{\rho}}{\omega^k - \bar{\rho}} \right)$$

now this is equal to

$$S_n = \frac{-1}{\rho - \bar{\rho}} \left[\rho \sum_{k=1}^n \frac{1}{\rho - \omega^k} - \bar{\rho} \sum_{k=1}^n \frac{1}{\bar{\rho} - \omega^k} \right].$$

We now propose an important lemma that will help with the evaluation of this summation.

Lemma: For any complex number α such that $\alpha^n \neq 1$ we have

$$\sum_{k=1}^n \frac{1}{\alpha - \omega^k} = \frac{n\alpha^{n-1}}{\alpha^n - 1}$$

To prove this let $P(z) = z^n - 1$ which can be factored as

$$P(z) = \prod_{k=1}^n (z - \omega^k)$$

now taking the logarithm of both sides gives

$$\ln(P(z)) = \sum_{k=1}^n \ln(z - \omega^k)$$

differentiating both sides with respect to z gives

$$\frac{P'(z)}{P(z)} = \sum_{k=1}^n \frac{1}{z - \omega^k}$$

thus

$$\sum_{k=1}^n \frac{1}{z - \omega^k} = \frac{nz^{n-1}}{z^n - 1}$$

as desired.

Now

$$S_n = \frac{-1}{\rho - \bar{\rho}} \left[\rho \left(\frac{n\rho^{n-1}}{\rho^n - 1} \right) - \bar{\rho} \left(\frac{n\bar{\rho}^{n-1}}{\bar{\rho}^n - 1} \right) \right]$$

which simplifies to

$$S_n = \frac{-n}{\rho - \bar{\rho}} \left[\frac{\rho^n}{\rho^n - 1} - \frac{\bar{\rho}^n}{\bar{\rho}^n - 1} \right].$$

Now we take cases based on mod 3.

If $n \equiv 1 \pmod{3}$ then let $n = 3m + 1$ so $\rho^n = \rho^{3m+1} = (\rho^3)^m \rho = \rho$ thus we have

$$\frac{\rho}{\rho - 1} - \frac{\bar{\rho}}{\bar{\rho} - 1} = \frac{\rho(\bar{\rho} - 1) - \bar{\rho}(\rho - 1)}{(\rho - 1)(\bar{\rho} - 1)}.$$

The numerator becomes $\rho\bar{\rho} - \rho - \rho\bar{\rho} + \bar{\rho} = -(\rho - \bar{\rho})$ and the denominator becomes $\rho\bar{\rho} - \rho - \bar{\rho} + 1$ which by Vieta's becomes $1 - (-1) + 1 = 3$ thus our sum is

$$S_n = \frac{-n}{\rho - \bar{\rho}} \left(\frac{-(\rho - \bar{\rho})}{3} \right) = \frac{n}{3}.$$

If $n \equiv 2 \pmod{3}$ then let $n = 3m + 2$ so $\rho^n = \rho^2 = \bar{\rho}$ and $\bar{\rho}^n = \bar{\rho}^2 = \rho$ so we have

$$\frac{\bar{\rho}}{\bar{\rho} - 1} - \frac{\rho}{\rho - 1}$$

which is the negative of what we had in the previous case thus this sum is

$$S_n = \frac{-n}{\rho - \bar{\rho}} \left(\frac{\rho - \bar{\rho}}{3} \right) = -\frac{n}{3}.$$

Note that then we have

$$S_{3m+1} + S_{3m+2} = \frac{3m+1}{3} - \frac{3m+2}{3} = -\frac{1}{3}$$

we have 33 such pairs that start from 1, 2 until 97, 98 so the sum of all these pairs is

$$33 \cdot \left(-\frac{1}{3} \right) = -11$$

but S_{100} hasn't been paired yet thus our sum is

$$S = -11 + \frac{100}{3} = -\frac{33}{3} + \frac{100}{3} = \frac{67}{3} \implies m + n = 67 + 3 = \boxed{70}.$$

proposed by Karam

Abhi wrote solutions to p1,p3,p5,p7,p9,p15 and Karam wrote solutions to the other 14 problems.

Div 1 Test B

1. The expression is built up by nesting, so we peel it off one layer at a time rather than expanding anything. Start from the outside:

$$((x+1)^2 + 1)^2 + 1 = 290 \implies ((x+1)^2 + 1)^2 = 289.$$

Taking square roots gives $(x+1)^2 + 1 = \pm 17$, but $(x+1)^2 + 1 \geq 1 > 0$, so we keep only the positive root:

$$(x+1)^2 + 1 = 17 \implies (x+1)^2 = 16 \implies x+1 = \pm 4.$$

That leaves $x = 3$ or $x = -5$. The condition $x < 0$ is exactly what picks between them (this is why the problem included it), so $x = -5$.

Now evaluate the second expression at $x = -5$, again from the inside out:

$$(x-1)^2 = (-6)^2 = 36, \quad 36-1 = 35, \quad 35^2 = 1225, \quad 1225-1 = 1224.$$

The last three digits are $\boxed{224}$.

Proposed by Shining

2. **Setting up coordinates.** The sheet has area 144, so it is 12×12 ; put it at $[0, 12]^2$. Fold along $y = 6$ to get the rectangle $[0, 12] \times [0, 6]$, then fold along $x = 6$ to get the square $[0, 6]^2$. Track the center of the original sheet, $(6, 6)$: after both folds it sits at the point $(6, 6)$, which is a *corner* of the folded square. Better still, the two edges meeting at that corner are precisely the two creases. This is the whole reason the problem works, and it also tells us which corner gets cut — if we snipped any other corner the holes would not land in the center.

What the cut unfolds into. A straight cut removing the corner $(6, 6)$ must meet the two edges through that corner, say at $(6-a, 6)$ and $(6, 6-b)$ with $a, b > 0$. It slices through all four layers. Unfolding reflects that little right triangle across both creases, so the hole is the quadrilateral with vertices

$$(6 \pm a, 6) \quad \text{and} \quad (6, 6 \pm b),$$

a rhombus centered at $(6, 6)$ with diagonals $2a$ and $2b$. Each of its four sides is a copy of the cut, of length $\sqrt{a^2 + b^2}$.

Using the perimeter. The perimeter condition says

$$4\sqrt{a^2 + b^2} = 24 \implies a^2 + b^2 = 36.$$

The hole's area, from the diagonal formula for a rhombus, is

$$[\text{hole}] = \frac{1}{2}(2a)(2b) = 2ab.$$

By AM-GM, $2ab \leq a^2 + b^2 = 36$, with equality when $a = b = 3\sqrt{2}$ (which comfortably fits inside the 6×6 square). So the hole can be as large as 36, and the paper left behind is

$$144 - 36 = \boxed{108}.$$

Proposed by Karam

3. Since $x - 2 \geq 0$, we must have $x \geq 2$. Move one radical to the other side to get $\sqrt{2x+5} = 3 + \sqrt{x-2}$. Then, square both sides to get $x - 2 = 6\sqrt{x-2}$. Let $y = \sqrt{x-2} \geq 0$. Then $y^2 = 6y$ so $y(y-6) = 0$. Hence, $y = 0 \implies x = 2$, or $y = 6 \implies x = 38$. Therefore, the sum of all solutions is $2 + 38 = 40$.

40

proposed by Karam

4. Write the number as $d_1d_2d_3d_4d_5$. The conditions are

$$d_1 < d_2 < d_3 \quad \text{and} \quad d_3 > d_4 > d_5.$$

Notice that d_3 appears in both chains — it is the peak of the number — so the natural move is to condition on it. Once $d_3 = k$ is fixed, the left half and the right half are chosen completely independently.

The left half. We need $d_1 < d_2 < k$, and since d_1 is the leading digit it cannot be 0. So d_1, d_2 is any 2-element subset of $\{1, 2, \dots, k-1\}$ (a set determines the order, since the digits must increase). That is $\binom{k-1}{2}$ ways.

The right half. We need $k > d_4 > d_5$, and here 0 is allowed, so d_4, d_5 is any 2-element subset of $\{0, 1, \dots, k-1\}$: that is $\binom{k}{2}$ ways.

The asymmetry between the two halves is exactly the leading-digit restriction, and it is the easiest thing to lose. Multiplying and summing over the peak k (which must be at least 3 for the left half to be possible):

$$N = \sum_{k=3}^9 \binom{k-1}{2} \binom{k}{2}.$$

Tabulating:

k	$\binom{k-1}{2}$	$\binom{k}{2}$	product
3	1	3	3
4	3	6	18
5	6	10	60
6	10	15	150
7	15	21	315
8	21	28	588
9	28	36	1008

Adding these up:

$$3 + 18 + 60 + 150 + 315 + 588 + 1008 = 2142.$$

So $N = 2142$, and its last three digits are 142.

Proposed by Abhi

5. Let $a = xy$, $b = yz$, $c = zx$. Then the system becomes

$$\begin{aligned} a + c &= 125, \\ a + b &= 89, \\ b + c &= 164. \end{aligned}$$

Adding the three equations gives $2(a + b + c) = 125 + 89 + 164 = 378$, so $a + b + c = 189$. Hence, $a = 189 - 164 = 25$, $b = 189 - 125 = 64$, $c = 189 - 89 = 100$. So, $xy = 25$, $yz = 64$, $zx = 100$. Multiplying these equations gives $(xy)(yz)(zx) = x^2y^2z^2 = (xyz)^2 = 25 \cdot 64 \cdot 100 = 160000$. Since $x, y, z > 0$,

$$xyz = \sqrt{160000} = \boxed{400}$$

Proposed by Karam

6. Let the 10 identical packages be distributed into 4 distinct mailboxes. By the Stars and Bars, the total number of equally likely ways to distribute the packages is

$$\text{Total Ways} = \binom{10+4-1}{4-1} = \binom{13}{3} = \frac{13 \times 12 \times 11}{3 \times 2 \times 1} = 286$$

By Linearity of Expectation, the expected number of overloaded mailboxes E is equal to 4 times the probability that a specific mailbox is overloaded namely

$$E = 4 \times P(\text{Mailbox 1 contains } \geq 4 \text{ packages})$$

To find the number of ways a specific mailbox receives 4 or more packages, we pre-allocate 4 packages to that mailbox. We then distribute the remaining $10 - 4 = 6$ packages across all 4 mailboxes without restriction

$$\text{Ways to Overload} = \binom{6+4-1}{4-1} = \binom{9}{3} = \frac{9 \times 8 \times 7}{3 \times 2 \times 1} = 84$$

So probability that a specific mailbox is overloaded is:

$$P = \frac{84}{286} = \frac{42}{143}$$

Now, we calculate the expected value E to be

$$E = 4 \times \frac{42}{143} = \frac{168}{143} \implies m + n = 168 + 143 = \boxed{311}$$

Proposed by Karam.

7. Let x be a k -digit positive integer, then $10^{k-1} \leq x < 10^k$. The maximum possible sum of digits for a k -digit number occurs when all digits are 9, which gives $f(x) \leq 9k$. Substituting this upper bound into the inequality yields

$$(9k)^2 \geq 10^{k-1} \implies 81k^2 \geq 10^{k-1}$$

We test small positive integers for k :

- For $k = 4$: $81(4)^2 = 1296 \geq 1000$ (Possible)
- For $k = 5$: $81(5)^2 = 2025 < 10000$ (Impossible)

Thus, x can have at most 4 digits namely $x \leq 9999$. If x is a 4-digit number, then the maximum possible value for its digit sum is $f(x) \leq 4 \times 9 = 36$. This implies

$$x \leq (f(x))^2 \leq 36^2 = 1296$$

Since $1000 \leq x \leq 1296$, the thousands digit of x must be 1. The maximum possible digit sum for a number in the range $[1000, 1296]$ occurs at $x = 1199$, giving $f(1199) = 1 + 1 + 9 + 9 = 20$. If $f(x) \leq 20$, then $(f(x))^2 \leq 20^2 = 400$. Since $x \geq 1000$, we would require $400 \geq 1000$, which is a contradiction. Therefore, no 4-digit solutions exist. Since there are no 4-digit solutions, we search for the largest 3-digit number ($100 \leq x \leq 999$). For any 3-digit number, $f(x) \leq 27$, so $x \leq 27^2 = 729$. To maximize x , we test values descending from the maximum possible digit sums

- **If $f(x) = 22$:** Then $(f(x))^2 = 22^2 = 484$. The smallest 3-digit number with a digit sum of 22 is 499 ($4 + 9 + 9 = 22$). Since $499 > 484$, the inequality fails.

- **If $f(x) = 21$:** Then $(f(x))^2 = 21^2 = 441$. We look for the largest 3-digit number $x \leq 441$ that satisfies $f(x) = 21$. Testing $x = 399$

$$f(399) = 3 + 9 + 9 = 21$$

Checking the original condition we see

$$(f(399))^2 = 21^2 = 441 \geq 399$$

The condition holds and any integer greater than 399 up to 441 will result in a significantly lower digit sum, failing the inequality. So the answer is

$$\boxed{399}.$$

Proposed by Shining

8. Since $g(x) \geq 0$, solutions can only occur where $\sin(10x) \geq 0$. On $[0, \pi]$, $\sin(10x)$ is positive on the five intervals

$$\left(0, \frac{\pi}{10}\right), \left(\frac{\pi}{5}, \frac{3\pi}{10}\right), \left(\frac{2\pi}{5}, \frac{\pi}{2}\right), \left(\frac{3\pi}{5}, \frac{7\pi}{10}\right), \left(\frac{4\pi}{5}, \frac{9\pi}{10}\right).$$

At each endpoint, $f(x) = 0 < g(x)$, while at the midpoint of each interval, $f(x) = 1 > g(x)$. Since $f(x) - g(x)$ is continuous and changes sign exactly once on each half of every positive interval, each interval contributes exactly two solutions. Thus, there are 10 solutions in $(0, \pi)$, together with the endpoint $x = \pi$, giving 11 solutions on $[0, \pi]$. Similarly, on $[-\pi, 0]$, there are four full positive intervals contributing 8 solutions and the partial interval $[-\pi, -9\pi/10]$, which contributes one interior solution together with the endpoint $x = -\pi$, for a total of 10 solutions.

$$11 + 10 = \boxed{21}$$

proposed by Karam

9. As always with these games, we hunt for the *losing positions*: the values of n such that the player *about to move* with n coins loses under optimal play. Call these L -positions.

By definition $n = 1$ is an L -position. Let us compute a few more by hand, remembering that from n coins you may leave anywhere from $\lceil n/2 \rceil$ to $n - 1$ coins.

- $n = 2$: take 1, leaving your opponent at 1. Win.
- $n = 3$: you may only take 1 (half of 3 rounds down), leaving 2 — a winning position for your opponent. Lose.
- $n = 4, 5, 6$: you can leave exactly 3 (take 1, 2, or 3 respectively, all within the half-limit). Win.
- $n = 7$: you may take at most 3, so you leave 4, 5, or 6 — all winners for your opponent. Lose.

The pattern 1, 3, 7, ... is unmistakable: the L -positions are $n = 2^k - 1$. Here is the proof, and both halves matter.

From $2^k - 1$ you are stuck. You may leave anything in $[2^{k-1}, 2^k - 2]$. The largest number of the form $2^j - 1$ in that window is $2^{k-1} - 1$, which is strictly below it, so every legal move lands you strictly between two consecutive L -positions — never on one.

From any other n you can escape. Let $2^k - 1$ be the largest such number less than n , so $2^k - 1 < n < 2^{k+1} - 1$, i.e. $n \leq 2^{k+1} - 2$. You want to remove $n - (2^k - 1)$ coins, which is at least 1; it is legal provided

$$n - (2^k - 1) \leq \frac{n}{2} \iff \frac{n}{2} \leq 2^k - 1 \iff n \leq 2^{k+1} - 2,$$

which is exactly the range we are in. So the move to $2^k - 1$ is always available.

Therefore the L -positions are precisely 1, 3, 7, 15, 31, 63, 127, 255, ..., and Shakira's job is to hand Pitbull one of them. Since $127 < 200 < 255$, she should leave 127 coins, which means removing

$$200 - 127 = \boxed{73}$$

coins. This is legal because $73 \leq 100$. From there she simply keeps returning Pitbull to 63, 31, 15, 7, 3, and finally 1.

Proposed by Val.

10. By Vieta's formulas, the sum of the roots is

$$a + b + c = 3$$

Thus, we can express the denominators as

$$b + c = 3 - a, \quad a + c = 3 - b, \quad a + b = 3 - c$$

The target sum simplifies to

$$S = \sum_{\text{cyc}} \frac{a^5}{3-a} = \frac{a^5}{3-a} + \frac{b^5}{3-b} + \frac{c^5}{3-c}$$

To simplify each term of the form $\frac{x^5}{3-x}$, let $u = 3 - x$, which implies $x = 3 - u$. We substitute $x = 3 - u$ into the original cubic equation to find the polynomial equation satisfied by u

$$(3-u)^3 - 3(3-u)^2 + 10(3-u) - 3 = 0$$

Expanding we get

$$(27 - 27u + 9u^2 - u^3) - 3(9 - 6u + u^2) + (30 - 10u) - 3 = 0$$

Simplifying we arrive at

$$-u^3 + 6u^2 - 19u + 27 = 0 \implies u^3 - 6u^2 + 19u - 27 = 0$$

We can now find the symmetric sums of u :

- $\sum u = 6$
- $\sum u_i u_j = 19$
- $\sum u^2 = (\sum u)^2 - 2 \sum u_i u_j = 6^2 - 2(19) = -2$

Now express $\frac{x^5}{3-x}$ completely in terms of u

$$\begin{aligned} \frac{x^5}{3-x} &= \frac{(3-u)^5}{u} = \frac{243 - 405u + 270u^2 - 90u^3 + 15u^4 - u^5}{u} \\ &= \frac{243}{u} - 405 + 270u - 90u^2 + 15u^3 - u^4 \end{aligned}$$

We use the relations from the cubic equation $u^3 - 6u^2 + 19u - 27 = 0$ to reduce the degree

1. $u^3 - 6u^2 + 19u = 27 \implies \frac{27}{u} = u^2 - 6u + 19 \implies \frac{243}{u} = 9u^2 - 54u + 171$
2. $u^4 = 6u^3 - 19u^2 + 27u$

Substituting these back yields

$$\begin{aligned} &= (9u^2 - 54u + 171) - 405 + 270u - 90u^2 + 15u^3 - (6u^3 - 19u^2 + 27u) \\ &= 9u^3 - 62u^2 + 189u - 234 \end{aligned}$$

Replacing u^3 with $6u^2 - 19u + 27$ gives

$$= 9(6u^2 - 19u + 27) - 62u^2 + 189u - 234$$

$$= -8u^2 + 18u + 9$$

We can now calculate the final total sum S which is

$$S = \sum_{i=1}^3 (-8u_i^2 + 18u_i + 9) = -8 \sum u_i^2 + 18 \sum u_i + 27$$

Plugging in our Vieta sums for u yields

$$S = -8(-2) + 18(6) + 27$$

$$S = 16 + 108 + 27 = \boxed{151}$$

Proposed by Shining+Abhi

11. Since $7 \mid d$, we must have $7 \mid n$ and $7 \mid 2^n - 1$. The order of 2 modulo 7 is 3, so $7 \mid 2^n - 1 \iff 3 \mid n$, so $21 \mid n$. Now we must ensure that 3 does not divide $2^n - 1$. Since $2^n - 1 \equiv (-1)^n - 1 \pmod{3}$, we have $3 \mid (2^n - 1) \iff n$ is even. Thus, n must be an odd multiple of 21. Finally, since the order of 2 modulo 5 is 4, $5 \mid 2^n - 1 \iff 4 \mid n$. Every multiple of $84 = \text{lcm}(21, 4)$ is even, so no odd multiple of 21 has $5 \mid 2^n - 1$. Therefore, the desired integers are odd multiples of 21 not exceeding 1000. There are $\lfloor \frac{1000}{21} \rfloor = 47$ multiples of 21, of which 24 are odd multiples. Hence, the answer is

$$\boxed{24}$$

proposed by Karam

12. Let $x = \log_a(b)$ and $y = \log_b(c)$. Using the change of base property, we know that $\log_a(c) = \log_a(b) \cdot \log_b(c) = xy$. We can translate the given equation $\log_a(b) + \log_b(c) = \log_a(c) = 2^k$ into a system of equations namely

$$\begin{aligned} x + y &= 2^k \\ xy &= 2^k \end{aligned}$$

Because we have the sum and product of x and y , they must be the roots of the quadratic equation

$$t^2 - 2^k t + 2^k = 0$$

Applying the quadratic formula gives

$$t = \frac{2^k \pm \sqrt{2^{2k} - 4(2^k)}}{2} = 2^{k-1} \pm \sqrt{2^{2k-2} - 2^k}$$

We are given that a, b , and c are positive integers plus $a \geq 2$ and $b \geq 2$. Because $x = \log_a(b)$, we can rewrite this as $b = a^x$. Notice that x is a root of a polynomial with integer coefficients ($t^2 - 2^k t + 2^k = 0$), which makes x an algebraic number. By Gelfond-Schneider theorem, if x were an irrational algebraic number, a^x would be a transcendental number however, we are given that b is an integer, meaning b must be algebraic this creates a contradiction and proves that x must be a rational number. For the root x to be rational, the expression inside the square root (the discriminant) must be a perfect square namely

$$2^{2k-2} - 2^k = 2^k(2^{k-2} - 1) = w^2$$

for some integer $w \geq 0$. Let's analyze the possible values for k

- If $k = 1$: The discriminant evaluates to $2^1(2^{-1} - 1) = -1$, yielding complex roots.
- If $k = 2$: The discriminant evaluates to $2^2(2^0 - 1) = 0$, which is a perfect square. This yields a single, rational double-root.

- If $k \geq 3$:

Case A (Even k): Let $k = 2m$ where $m \geq 2$. For $2^{2m}(2^{2m-2} - 1)$ to be a perfect square, the odd term $2^{2m-2} - 1$ must be a perfect square. The only perfect squares that differ by 1 are 0 and 1, meaning $2^{2m-2} - 1 = 0 \implies m = 1 \implies k = 2$. This contradicts $k \geq 3$.

Case B (Odd k): Let $k = 2m + 1$ where $m \geq 1$. The discriminant is $2^{2m+1}(2^{2m-1} - 1)$. For this to be a square, $2(2^{2m-1} - 1)$ must be a square, which requires $2^{2m-1} - 1 = 2r^2$ for some integer r . If $m = 1$, $2^1 - 1 = 1$, which does not equal $2r^2$. If $m \geq 2$, 2^{2m-1} is a multiple of 8, meaning $2^{2m-1} - 1 \equiv -1 \equiv 7 \pmod{8}$. However, $2r^2 \pmod{8}$ can only ever be 0 or 2. Thus, there are no solutions for odd k .

This proves that $k = 2$ is the unique solution that satisfies the conditions. For $k = 2$, the double root of our quadratic is $x = y = 2$. We need to evaluate $\log_{ab}(c) - 1$ we get

$$\log_{ab}(c) = \frac{\log_a(c)}{\log_a(ab)} = \frac{\log_a(b) \cdot \log_b(c)}{\log_a(a) + \log_a(b)} = \frac{xy}{1+x}$$

Substituting $x = 2$ and $y = 2$ yields

$$\log_{ab}(c) = \frac{2 \cdot 2}{1 + 2} = \frac{4}{3}$$

Therefore the expression that we want is

$$\log_{ab}(c) - 1 = \frac{4}{3} - 1 = \frac{1}{3} \implies 1 + 3 = \boxed{4}$$

Proposed by Shining

13. Let the angles of $\triangle ABC$ be α, β , and γ . Because AD is an altitude, $\triangle ABD$ and $\triangle ACD$ are right triangles. We can easily find the side lengths of $\triangle ABC$, in $\triangle ABD$: $AB = \sqrt{AD^2 + BD^2} = \sqrt{24^2 + 10^2} = 26$. In $\triangle ACD$: $AC = \sqrt{AD^2 + CD^2} = \sqrt{24^2 + 18^2} = 30$. Now, we can find the sine and cosine of angles β and γ

- $\sin \beta = \frac{AD}{AB} = \frac{24}{26} = \frac{12}{13}$
- $\cos \beta = \frac{BD}{AB} = \frac{10}{26} = \frac{5}{13}$
- $\sin \gamma = \frac{AD}{AC} = \frac{24}{30} = \frac{4}{5}$
- $\cos \gamma = \frac{CD}{AC} = \frac{18}{30} = \frac{3}{5}$

Since $\alpha = 180^\circ - (\beta + \gamma)$, we can find $\cos \alpha$ to be

$$\begin{aligned} \cos \alpha &= -\cos(\beta + \gamma) = -(\cos \beta \cos \gamma - \sin \beta \sin \gamma) \\ \cos \alpha &= -\left(\frac{5}{13} \cdot \frac{3}{5} - \frac{12}{13} \cdot \frac{4}{5}\right) = -\left(\frac{15}{65} - \frac{48}{65}\right) = \frac{33}{65} \end{aligned}$$

Circle ω has diameter $AD = 24$, so its center O is the midpoint of AD , and its radius is $R = 12$. Because $\angle AED$ and $\angle AFD$ are inscribed angles that subtend the diameter AD , they must be 90° . This means E and F are simply the feet of the altitudes from D to AB and AC . Furthermore, P is the intersection of the tangents to ω at E and F . Namely, P is the pole of chord EF with respect to ω . In $\triangle OEF$, $OE = OF = R = 12$. The line OP is the perpendicular bisector of EF , meaning OP bisects $\angle EOF$. Since PE is tangent to ω we have $\angle OEP = 90^\circ$. First, we find the central angles of ω : $\triangle OAE$ is isosceles ($OA = OE$), so $\angle OAE = \angle BAD = 90^\circ - \beta$. The exterior angle at O is $\angle EOA = 180^\circ - 2(90^\circ - \beta) = 2\beta$. By identical logic on the other side, $\angle FOA = 2\gamma$. The total angle subtended by the major arc EF (the one containing A) is $\angle EOA + \angle FOA = 2\beta + 2\gamma$. The minor arc EF (not containing A) subtends an angle of $360^\circ - (2\beta + 2\gamma) = 2(180^\circ - (\beta + \gamma)) = 2\alpha$. Since OP bisects the minor arc EF , we have $\angle EOP = \alpha$. In the right triangle $\triangle OEP$ we have

$$\cos(\angle EOP) = \frac{OE}{OP} \implies \cos \alpha = \frac{12}{OP}$$

$$OP = \frac{12}{\cos \alpha} = \frac{12}{33/65} = \frac{12 \cdot 65}{33} = \frac{260}{11}$$

To find the distance AP , we will use the Law of Cosines on $\triangle AOP$. We need the angle between the ray OA and the ray OP . Imagine looking from O towards A (let's call this 0°). Ray OE is at an angle of 2β to one side, Ray OF is at an angle of 2γ to the other side. The ray bisecting the major arc EF would be exactly halfway between them, at an angle of $\frac{2\beta-2\gamma}{2} = \beta - \gamma$ relative to OA . Since OP bisects the minor arc EF , it points in the exact opposite direction of the major arc bisector therefore, the angle $\angle AOP$ is $180^\circ - (\beta - \gamma)$. We now find it's cosine

$$\begin{aligned}\cos(\angle AOP) &= \cos(180^\circ - (\beta - \gamma)) = -\cos(\beta - \gamma) \\ \cos(\beta - \gamma) &= \cos \beta \cos \gamma + \sin \beta \sin \gamma = \left(\frac{5}{13}\right)\left(\frac{3}{5}\right) + \left(\frac{12}{13}\right)\left(\frac{4}{5}\right) = \frac{15}{65} + \frac{48}{65} = \frac{63}{65} \\ \cos(\angle AOP) &= -\frac{63}{65}\end{aligned}$$

Now we apply the Law of Cosines to $\triangle AOP$ to get

$$AP^2 = OA^2 + OP^2 - 2(OA)(OP) \cos(\angle AOP)$$

$$AP^2 = 12^2 + \left(\frac{260}{11}\right)^2 - 2(12)\left(\frac{260}{11}\right)\left(-\frac{63}{65}\right)$$

the last term is $2 \cdot 12 \cdot \frac{260}{11} \cdot \frac{63}{65} = 24 \cdot \frac{4}{11} \cdot 63 = \frac{6048}{11}$.

$$AP^2 = 144 + \frac{67600}{121} + \frac{6048}{11}$$

now we add everything up

$$\begin{aligned}AP^2 &= \frac{17424}{121} + \frac{67600}{121} + \frac{66528}{121} = \frac{151552}{121} \\ AP &= \frac{\sqrt{151552}}{11} = \frac{64\sqrt{37}}{11} \implies a + b + c = 64 + 37 + 11 = \boxed{112}\end{aligned}$$

Proposed by Karam

14. We can interpret this algebraic system geometrically using the Law of Cosines. For a triangle with side lengths A and B that meet at an angle of 120° , the length of the opposite side C satisfies

$$C^2 = A^2 + B^2 - 2AB \cos(120^\circ) = A^2 + AB + B^2$$

Thus, the system of equations describes a point P inside an outer triangle $\triangle A_1A_2A_3$ such that the segments connecting P to the three vertices have lengths x , y , and z , and all three central angles meeting at P are exactly 120° . This configuration means P is the Fermat-Torricelli point of the outer triangle. The side lengths of the outer triangle $\triangle A_1A_2A_3$ are given by the square roots namely

- $c = \sqrt{x^2 + xy + y^2} = \sqrt{40}$
- $a = \sqrt{y^2 + yz + z^2} = \sqrt{25} = 5$
- $b = \sqrt{z^2 + zx + x^2} = \sqrt{49} = 7$

The total area of the outer triangle is the sum of the areas of the three internal triangles meeting at P , each having a vertex angle of 120° :

$$\begin{aligned}\text{Area} &= \frac{1}{2}xy \sin(120^\circ) + \frac{1}{2}yz \sin(120^\circ) + \frac{1}{2}zx \sin(120^\circ) \\ \text{Area} &= \frac{\sqrt{3}}{4}(xy + yz + zx) = \frac{\sqrt{3}}{4}S\end{aligned}$$

We rewrite this relation to isolate our target value S

$$S = \frac{4}{\sqrt{3}} \times \text{Area}$$

Next, we calculate the area of the outer triangle with side lengths 5, 7, and $\sqrt{40}$ using Heron's formula. The semiperimeter s is

$$s = \frac{5 + 7 + \sqrt{40}}{2} = 6 + \sqrt{10}$$

Applying Heron's gives

$$\begin{aligned} \text{Area} &= \sqrt{s(s-a)(s-b)(s-c)} \\ \text{Area} &= \sqrt{(6 + \sqrt{10}) \cdot (6 + \sqrt{10} - 5) \cdot (6 + \sqrt{10} - 7) \cdot (6 + \sqrt{10} - \sqrt{40})} \\ \text{Area} &= \sqrt{(6 + \sqrt{10})(1 + \sqrt{10})(-1 + \sqrt{10})(6 - \sqrt{10})} \end{aligned}$$

Grouping the conjugate pairs gives

$$\begin{aligned} \text{Area} &= \sqrt{[(6 + \sqrt{10})(6 - \sqrt{10})] \cdot [(\sqrt{10} + 1)(\sqrt{10} - 1)]} \\ \text{Area} &= \sqrt{(36 - 10)(10 - 1)} = \sqrt{26 \times 9} = 3\sqrt{26} \end{aligned}$$

Finally, we substitute the area back into the expression for S to arrive at

$$S = \frac{4}{\sqrt{3}} \cdot 3\sqrt{26} = 4\sqrt{3} \cdot \sqrt{26} = 4\sqrt{78} \implies a + b = 4 + 78 = \boxed{82}.$$

Proposed by Karam

15. Let the total number of paths of length N that return to the origin be W_N which only works when N is even so let $N = 2n$. Note that we need an equal number of right steps, left steps and an equal number of up steps, down steps so we have

$$W_{2n} = \binom{2n}{n}^2$$

such paths. Let f_{2k} be the number of paths of length $2k$ where the walk returns to the origin at step $2k$ for the first time. If the first return to $(0,0)$ occurs at step $2k$ (where $2 \leq 2k \leq 2n$) then we have the remainder of the path is simply a path of length $2n - 2k$ starting and ending at the origin. So we arrive at the following

$$W_{2n} = \sum_{k=1}^n f_{2k} W_{2n-2k}$$

see that from this we get

$$f_{2n} = W_{2n} - \sum_{k=1}^{n-1} f_{2k} W_{2n-2k}.$$

Now $W_2 = \binom{2}{1}^2 = 4$ and $W_4 = \binom{4}{2}^2 = 36$ and $W_6 = \binom{6}{3}^2 = 400$ and $W_8 = \binom{8}{4}^2 = 4900$ and $W_{10} = \binom{10}{5}^2 = 63504$. Again these calculations look nasty but taking modulo 1000 makes it way simpler. See that f_2 : $W_2 = 4$ and f_4 : $W_4 - f_2 W_2 = 36 - (4)(4) = 20$ and f_6 : $W_6 - (f_2 W_4 + f_4 W_2) = 400 - (4 \cdot 36 + 20 \cdot 4) = 400 - 224 = 176$ and f_8 : $W_8 - (f_2 W_6 + f_4 W_4 + f_6 W_2) = 4900 - (4 \cdot 400 + 20 \cdot 36 + 176 \cdot 4) = 4900 - 3024 = 1876$ and finally we compute f_{10} :

$$W_{10} - (f_2 W_8 + f_4 W_6 + f_6 W_4 + f_8 W_2) = 63504 - (4 \cdot 4900 + 20 \cdot 400 + 176 \cdot 36 + 1876 \cdot 4) = 63504 - 41440 = 22064$$

taking modulo 1000 gives

$$\boxed{64}.$$

proposed by Karam

16. Let the original number N have k digits. We can separate N into its last digit, d , and the remaining part at the front which we will call x . See that $0 \leq d \leq 9$ and because N is a positive integer and we are moving d to the front to create another integer that is larger ($\frac{13}{4}N$), d cannot be 0. We can express N algebraically as $N = 10x + d$. Since N has k digits, x must have $k - 1$ digits. When we move the rightmost digit d to the very front, the new number N' is formed by shifting d by $k - 1$ place values namely

$$N' = d \cdot 10^{k-1} + x$$

The problem states that $N' = \frac{13}{4}N$. To avoid fractions, we write this as

$$4N' = 13N$$

Substituting yields

$$4(d \cdot 10^{k-1} + x) = 13(10x + d)$$

$$4d \cdot 10^{k-1} + 4x = 130x + 13d$$

Isolate the terms with x on one side and d on the other to get

$$4d \cdot 10^{k-1} - 13d = 126x$$

$$d(4 \cdot 10^{k-1} - 13) = 126x$$

Solving for x , we get

$$x = \frac{d(4 \cdot 10^{k-1} - 13)}{126}$$

For x to be an integer, the numerator $d(4 \cdot 10^{k-1} - 13)$ must be divisible by 126. The prime factorization of 126 is $126 = 2 \cdot 9 \cdot 7$ so we must test the divisibility of our numerator with 2, 9, and 7. We first check modulo 2 (which is just parity). For any $k \geq 2$, 10^{k-1} is even, so $4 \cdot 10^{k-1}$ is even, subtracting 13 makes the term $(4 \cdot 10^{k-1} - 13)$ odd. Because 126 is even, the factor of 2 must come from d thus d must be an even digit

$$d \in \{2, 4, 6, 8\}$$

Let's check the odd term modulo 9. We know $10 \equiv 1 \pmod{9}$, so $10^{k-1} \equiv 1 \pmod{9}$. So we have

$$4(1) - 13 = -9 \equiv 0 \pmod{9}$$

So $(4 \cdot 10^{k-1} - 13)$ is always divisible by 9, regardless of the value of k . We need the remaining factor of 7 to divide $d(4 \cdot 10^{k-1} - 13)$. Since $d \in \{2, 4, 6, 8\}$, d is never divisible by 7. Thus, we must have

$$4 \cdot 10^{k-1} - 13 \equiv 0 \pmod{7}$$

We can rewrite this as

$$4 \cdot 3^{k-1} + 1 \equiv 0 \pmod{7}$$

$$4 \cdot 3^{k-1} \equiv -1 \equiv 6 \pmod{7}$$

Multiply both sides by 2 to get

$$3^{k-1} \equiv 12 \equiv 5 \pmod{7}$$

Now, let's look at the powers of 3 $\pmod{7}$ they are

- $3^1 \equiv 3$
- $3^2 \equiv 2$
- $3^3 \equiv 6$
- $3^4 \equiv 4$

- $3^5 \equiv 5 \leftarrow$ this is our target.
- $3^6 \equiv 1$

For $3^{k-1} \equiv 5 \pmod{7}$, the exponent must satisfy $k-1 \equiv 5 \pmod{6}$, which simplifies to

$$k \equiv 0 \pmod{6}$$

The number of digits in N , k , must be a multiple of 6. Let $k = 6m$ for some positive integer m . Let's plug $k = 6m$ back into our equation for N to get

$$N = 10x + d = 10 \left(\frac{d(4 \cdot 10^{6m-1} - 13)}{126} \right) + d$$

$$N = \frac{40d \cdot 10^{6m-1} - 130d + 126d}{126}$$

$$N = \frac{4d \cdot 10^{6m} - 4d}{126} = \frac{2d(10^{6m} - 1)}{63}$$

The original number N must have exactly $k = 6m$ digits. This means $10^{6m-1} \leq N < 10^{6m}$. Let's test our valid even digits $\{2, 4, 6, 8\}$ in our formula for N . If $d = 2$ then

$$N = \frac{4(10^{6m} - 1)}{63} \approx 0.0634 \times 10^{6m}$$

Because $0.0634 < 0.1$, N is strictly less than 10^{6m-1} . This means if $d = 2$, the resulting N will only have $6m - 1$ digits thus $d = 2$ is an extraneous root and is invalid. So the only valid digits are $d \in \{4, 6, 8\}$. Let $U_m = \frac{10^{6m} - 1}{63}$. For $m = 1$, $U_1 = \frac{999999}{63} = 15873$. For $m = 2$, $U_2 = \frac{9999999999}{63} = 15873015873$. In general, U_m is the block '015873' repeated m times (with the leading zero dropped on the very first block). For any given m , our valid integers N are

- $N_{m,1} = 2(4)U_m = 8U_m$
- $N_{m,2} = 2(6)U_m = 12U_m$
- $N_{m,3} = 2(8)U_m = 16U_m$

Because $U_1 < U_2 < U_3 \dots$, the infinite sequence $N_1 < N_2 < N_3 \dots$ is formed by taking $m = 1, 2, 3 \dots$ in groups of three

- $N_1 = 8U_1$
- $N_2 = 12U_1$
- $N_3 = 16U_1$
- $N_4 = 8U_2$
- ...

We are looking for the 25th term in this sequence. Since $25 = 3(8) + 1$, N_{25} corresponds to the first term of the 9th group ($m = 9$). See that

$$N_{25} = 8U_9$$

We need to evaluate $8U_9 \pmod{1000}$. Remember, the structure of U_m is a repeating block of 15873 namely

$$U_9 = 15873015873\dots015873$$

Regardless of how massive m gets, the last six digits of U_m will always be 015873 so the last three digits of U_9 are 873. Thus

$$U_9 \equiv 873 \pmod{1000}$$

$$N_{25} \equiv 8(873) \equiv 6984 \equiv \boxed{984} \pmod{1000}$$

Proposed by Phillip

17. Notice that $20^2 + 21^2 = 400 + 441 = 841 = 29^2$, which means $\triangle ABC$ is a right triangle with $\angle C = 90^\circ$. Let s be the semiperimeter and r be the inradius:

- $s = \frac{20+21+29}{2} = 35$
- $r = \frac{Area}{s} = \frac{\frac{1}{2}(20)(21)}{35} = \frac{210}{35} = 6$

Furthermore, we can find values for half-angle A . Since $\cos A = \frac{21}{29}$ we have

$$\cos\left(\frac{A}{2}\right) = \sqrt{\frac{1 + \cos A}{2}} = \sqrt{\frac{50}{58}} = \frac{5}{\sqrt{29}}$$

$$\sin\left(\frac{A}{2}\right) = \sqrt{\frac{1 - \cos A}{2}} = \sqrt{\frac{8}{58}} = \frac{2}{\sqrt{29}}$$

$$\tan\left(\frac{A}{2}\right) = \frac{2}{5}$$

Because $AE = AF$, the angle bisector AI is the perpendicular bisector of EF . Thus, $AI \perp EF$, by definition of the orthocenter, the altitude from D to EF passes through P , meaning $DP \perp EF$. Because both DP and AI are perpendicular to EF , they must be parallel so $DP \parallel AI$. By applying this exact same symmetry to the other vertices, we get

- $DP \parallel AI$
- $EP \parallel BI$
- $FP \parallel CI$

We are given a line ℓ that passes through I such that $\ell \perp AI$. The points X and Y are the intersections of ℓ with PE and PF , respectively. We compute the angle at X . Because PX is simply the line PE , we know $PX \parallel BI$ therefore, the angle $\angle PXY$ is equal to the angle between BI and ℓ . Since $\ell \perp AI$, this angle is $|90^\circ - \angle AIB|$. In $\triangle AIB$, $\angle AIB = 180^\circ - \frac{A+B}{2}$, because $\angle C = 90^\circ$, $A + B = 90^\circ$, meaning $\angle AIB = 135^\circ$. Thus, $\angle PXY = 135^\circ - 90^\circ = 45^\circ$. Now we find the angle at Y . Because PY is the line PF , $PY \parallel CI$. The angle $\angle PYX$ is the angle between CI and ℓ , which is $|90^\circ - \angle AIC|$. In $\triangle AIC$, $\angle AIC = 180^\circ - \frac{A+C}{2} = 180^\circ - (\frac{A}{2} + 45^\circ) = 135^\circ - \frac{A}{2}$. Thus, $\angle PYX = (135^\circ - \frac{A}{2}) - 90^\circ = 45^\circ - \frac{A}{2}$. Let h be the distance from P to the line ℓ . We can determine h by treating I as the origin and evaluating the vector projection of $\vec{IP}$ onto AI . For an inscribed triangle DEF with circumcenter I and orthocenter P see that

$$\vec{IP} = \vec{ID} + \vec{IE} + \vec{IF}$$

Let $\vec{u}$ be the unit vector in the direction of $\vec{IA}$, the line ℓ is orthogonal to $\vec{u}$, so $h = |\vec{IP} \cdot \vec{u}|$. We compute the component of $(\vec{IE} + \vec{IF})$ first. The sum of vectors $\vec{IE}$ and $\vec{IF}$ points exactly along $\vec{IA}$, because the angle between $\vec{IA}$ and $\vec{IE}$ is $90^\circ - \frac{A}{2}$, the magnitude of this combined vector is $2r \cos(90^\circ - \frac{A}{2}) = 2r \sin(\frac{A}{2})$. Now we compute the component of $\vec{ID}$. Because $\angle C = 90^\circ$, $ID \perp BC$ and $AC \perp BC$ means that $ID \parallel AC$ therefore, the angle between $\vec{ID}$ (pointing away from AC) and $\vec{IA}$ is $180^\circ - \frac{A}{2}$. Its projection onto $\vec{u}$ is $r \cos(180^\circ - \frac{A}{2}) = -r \cos(\frac{A}{2})$. Summing these projections, the total distance h to ℓ is

$$h = r \left| 2 \sin\left(\frac{A}{2}\right) - \cos\left(\frac{A}{2}\right) \right|$$

Substituting our known values gives

$$h = 6 \left| 2 \left(\frac{2}{\sqrt{29}} \right) - \left(\frac{5}{\sqrt{29}} \right) \right| = 6 \left| \frac{4-5}{\sqrt{29}} \right| = \frac{6}{\sqrt{29}}$$

We now have a triangle $\triangle PXY$ with height $h = \frac{6}{\sqrt{29}}$ and base angles $\angle X = 45^\circ$ and $\angle Y = 45^\circ - \frac{A}{2}$. Since both angles are acute, the altitude drops strictly between X and Y so

$$XY = h \cot X + h \cot Y$$

$$\text{Area}(PXY) = \frac{1}{2}(XY)h = \frac{1}{2}h^2(\cot X + \cot Y)$$

See that

$$\begin{aligned} \cot X &= \cot 45^\circ = 1 \\ \cot Y &= \cot \left(45^\circ - \frac{A}{2} \right) = \frac{1 + \tan(A/2)}{1 - \tan(A/2)} = \frac{1 + 2/5}{1 - 2/5} = \frac{7/5}{3/5} = \frac{7}{3} \end{aligned}$$

Finally, plug everything in

$$\begin{aligned} \text{Area}(PXY) &= \frac{1}{2} \left(\frac{36}{29} \right) \left(1 + \frac{7}{3} \right) \\ \text{Area}(PXY) &= \frac{18}{29} \left(\frac{10}{3} \right) = \frac{60}{29} \implies m + n = 60 + 29 = \boxed{89}. \end{aligned}$$

Proposed by Karam

18. The generating function tracking the subset sums of S is given by

$$P(x) = \prod_{k=1}^{17} (1 + x^k)$$

Let $\zeta = e^{2\pi i/5}$ be a primitive 5-th root of unity, by roots-of-unity filter theorem, the number of subsets N whose element sum is congruent to 0 (mod 5) is given by

$$N = \frac{1}{5} \sum_{j=0}^4 P(\zeta^j) \tag{1}$$

For $j = 0$, we substitute $x = \zeta^0 = 1$ into our generating function:

$$P(1) = \prod_{k=1}^{17} (1 + 1) = 2^{17} = 131072$$

For any $j \in \{1, 2, 3, 4\}$, $\omega = \zeta^j$ behaves as a primitive 5-th root of unity. The sequence of exponents k from 1 to 17 modulo 5 yields exactly three complete periods of $\{0, 1, 2, 3, 4\}$ (accounting for numbers 1 to 15) and a remainder group $\{1, 2\}$ (accounting for 16 and 17). First, we determine the product over one full period of roots of unity

$$\prod_{m=0}^4 (1 + \omega^m) = (1 + 1)(1 + \omega)(1 + \omega^2)(1 + \omega^3)(1 + \omega^4)$$

Using the identity $\prod_{m=1}^4 (x - \omega^m) = x^4 + x^3 + x^2 + x + 1$, we substitute $x = -1$ to get

$$\prod_{m=1}^4 (-1 - \omega^m) = (-1)^4 + (-1)^3 + (-1)^2 + (-1) + 1 = 1$$

Since there are an even number of factors, $(-1)^4 \prod_{m=1}^4 (1 + \omega^m) = \prod_{m=1}^4 (1 + \omega^m) = 1$. Thus, the product over a full period is $2 \cdot 1 = 2$. With three full periods and the two remaining terms, the expression for each primitive root becomes

$$P(\zeta^j) = 2^3 \cdot (1 + \zeta^j)(1 + \zeta^{2j}) = 8(1 + \zeta^j + \zeta^{2j} + \zeta^{3j})$$

Now, summing this across all $j = 1, 2, 3, 4$ yields

$$\sum_{j=1}^4 P(\zeta^j) = 8 \sum_{j=1}^4 (1 + \zeta^j + \zeta^{2j} + \zeta^{3j})$$

Using the basic property of roots of unity where $\sum_{j=1}^4 \zeta^{mj} = -1$ for any integer $m \not\equiv 0 \pmod{5}$ we get

$$\sum_{j=1}^4 P(\zeta^j) = 8 \left[4 + (-1) + (-1) + (-1) \right] = 8 \cdot 1 = 8$$

Combining our results gives

$$N = \frac{1}{5} \left(P(1) + \sum_{j=1}^4 P(\zeta^j) \right) = \frac{131072 + 8}{5} = \frac{131080}{5} = 26 \boxed{216}$$

Proposed by Abhi

19. Let $N = 4054 = 2 \times 2027$, where 2027 is a prime number so the given sum can be written as

$$S = \sum_{k=1}^{N^2} \gcd(k, N) \cdot k^E$$

where $E = E_{2026} = 2026^{E_{2025}}$. We wish to find the remainder $X = S \pmod{N^2}$. Any index k in the range $1 \leq k \leq N^2$ can be expressed as $k = qN + r$, where $0 \leq q < N$ and $1 \leq r \leq N$. The greatest common divisor depends only on the remainder r

$$\gcd(k, N) = \gcd(qN + r, N) = \gcd(r, N)$$

Using the binomial expansion, the power term k^E simplifies modulo N^2 to

$$k^E = (qN + r)^E \equiv r^E + E \cdot qNr^{E-1} \pmod{N^2}$$

Substituting these components into the expression for S yields

$$S \equiv \sum_{r=1}^N \sum_{q=0}^{N-1} \gcd(r, N) (r^E + E \cdot qNr^{E-1}) \pmod{N^2}$$

Splitting the inner summations over q gives us

1. For the first term: $\sum_{q=0}^{N-1} 1 = N$.
2. For the second term: $\sum_{q=0}^{N-1} q = \frac{N(N-1)}{2}$.

Since $E = 2026^{E_{2025}}$ is a massive power of an even base, E is an even integer. Thus, the second component contains the factor $E \cdot \frac{N^2(N-1)}{2} \equiv 0 \pmod{N^2}$, causing it to vanish entirely causing the sum to simplify to

$$S \equiv N \sum_{r=1}^N \gcd(r, N) r^E \pmod{N^2}$$

To determine $X = S \pmod{N^2}$, it is sufficient to evaluate the inner sum $A = \sum_{r=1}^N \gcd(r, N) r^E$ modulo $N = 4054$. We do this by checking it modulo 2 and modulo 2027. First we consider modulo 2.

- If r is even, then $\gcd(r, N)r^E \equiv 0 \pmod{2}$.
- If r is odd, then $\gcd(r, N)$ is odd and $r^E \equiv 1 \pmod{2}$.
- There are exactly 2027 odd integers in the range $1 \leq r \leq 4054$, so

$$A \equiv 2027 \times 1 \equiv 1 \pmod{2}$$

Now we consider modulo 2027.

- If $2027 \mid r$, the term vanishes modulo 2027.
- If $2027 \nmid r$, by Fermat's Little Theorem, $r^E \equiv 1 \pmod{2027}$ because E is a multiple of $\phi(2027) = 2026$.
- For these non-divisible elements, $\gcd(r, N) = 1$ if r is odd, and $\gcd(r, N) = 2$ if r is even.
- Excluding the multiples of 2027 (2027 and 4054), there remain 2026 odd numbers and 2026 even numbers yielding

$$A \equiv 2026 \times 1 + 2026 \times 2 = 3 \times 2026 \equiv 3 \times (-1) = -3 \equiv 2024 \pmod{2027}$$

Combining the congruences using the Chinese Remainder Theorem gives

$$A \equiv 1 \pmod{2} \quad \text{and} \quad A \equiv 2024 \pmod{2027}$$

Since 2024 is even, the solution in the range is

$$A \equiv 2024 + 2027 = 4051 \pmod{4054}$$

The actual remainder value X is

$$X = N \cdot (A \pmod{N}) = 4054 \times 4051$$

We compute this modulo 1000

$$X \equiv 54 \times 51 = 2754 \equiv \boxed{754} \pmod{1000}$$

Proposed by Karam

20. Since $x \neq 0$, we can divide both sides by x^5

$$x^5 + 1 + x^{-5} = \prod_{k=1}^5 (x + a_k + x^{-1})$$

Let $z = x + x^{-1}$. We can express $x^5 + x^{-5}$ as a polynomial in z using the standard expansion derived from Chebyshev polynomials

$$x^5 + x^{-5} = z^5 - 5z^3 + 5z$$

Substituting this relation into the equation defines a new polynomial $Q(z)$

$$Q(z) = z^5 - 5z^3 + 5z + 1 = \prod_{k=1}^5 (z + a_k)$$

Consider a polynomial $f(w)$ whose roots are exactly $w_k = a_k + 2$. Using the relation $\prod_{k=1}^5 (z + a_k) = Q(z)$, we set $z = -(w - 2) = 2 - w$ to get

$$f(w) = \prod_{k=1}^5 (w - (a_k + 2)) = (-1)^5 \prod_{k=1}^5 (2 - w + a_k) = -Q(2 - w)$$

For any polynomial $f(w) = c_5w^5 + c_4w^4 + \cdots + c_1w + c_0$, the sum of the reciprocals of its roots is determined by the linear and constant coefficients namely

$$\sum_{k=1}^5 \frac{1}{w_k} = -\frac{c_1}{c_0} = -\frac{f'(0)}{f(0)}$$

Differentiating $f(w) = -Q(2 - w)$ with respect to w using the chain rule gives

- $f(0) = -Q(2)$
- $f'(0) = Q'(2)$

Now, we evaluate $Q(2)$ and $Q'(2)$

$$Q(2) = 2^5 - 5(2^3) + 5(2) + 1 = 32 - 40 + 10 + 1 = 3$$

$$Q'(z) = 5z^4 - 15z^2 + 5 \implies Q'(2) = 5(16) - 15(4) + 5 = 80 - 60 + 5 = 25$$

Substituting these back gives us our answer

$$\sum_{k=1}^5 \frac{1}{a_k + 2} = -\frac{25}{-3} = \frac{25}{3} \implies 25 + 3 = \boxed{28}$$

Proposed by Neil

Solutions to p1,p2,p4,p9 by Bryan, solutions to p3,p5,p8,p11 by Vajpai, and solutions to p6,p7,p10,p12 through 20 by Karam.